American International University-Bangladesh (AIUB)

Department of Computer Science

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Agricultural Equipment Rental and Management System

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1. PROJECT PROPOSAL

1.1 Background to the Problem

In modern day farming, machinery like tractors, harvesters and other machinery are critical in improving productivity and reducing the amount of manual labor. However, small and medium scale farmers cannot afford such expensive machines, especially those working on small scale. Therefore, they either use common practices or informal rental agreements, which are often inefficient, time consuming and unreliable.

At the same time, there is not an efficient centralized place where farmers can easily find equipment they can hire on a need basis. Owners and service providers of equipment also face difficulty in managing equipment, availability tracking, booking and record keeping. In addition to this, an unstructured system poses difficulties to employees in organizing their operations like delivery, maintenance, and processing of rentals. As a result, such problems create problems with booking, lack of transparency, poor communication, and poor use of resources.

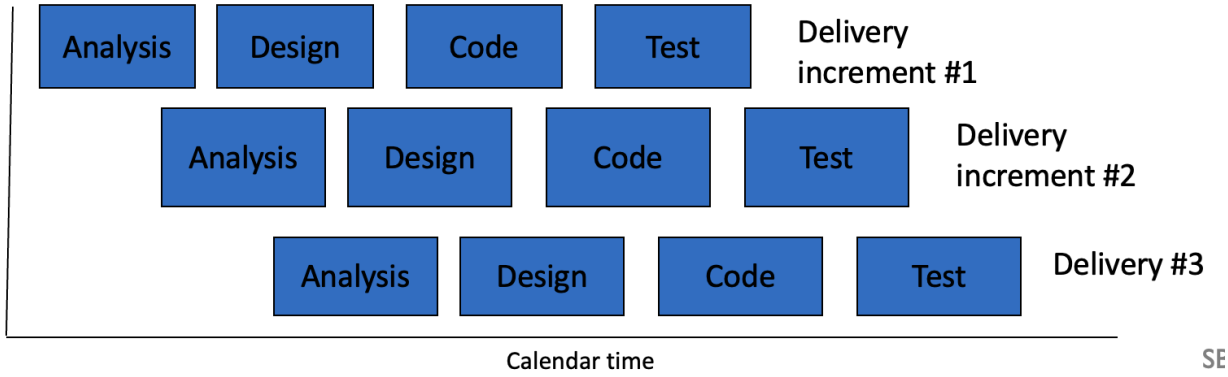
The proposed Agricultural Equipment Rental and Management System is capable of addressing these challenges by creating a digital hub, which would be centralized, and through which farmers can easily browse and hire equipment and employees manage operations and administrators supervise activities of the system. The system makes access more transparent, provides effective equipment control, reduces human input, and improves interaction with the user. It is therefore a good, efficient, and user-friendly solution to the development of the agricultural equipment rental services.

1.2 Selection of Process Model

For our **Agricultural Equipment Rental and Management System**, we selected the **Incremental Model**.

The Incremental Model is a software development process where the system is developed step by step in several increments or modules. Instead of developing the whole system at once, the project is divided into smaller functional parts. Each increment goes through analysis, design, development, testing, and delivery before moving to the next increment.

In our project, the system is divided into multiple modules such as Authentication, Equipment Management, Booking and Payment, Support and Smart Features, and Admin & Maintenance. Each module is developed and tested separately. After completing one increment, the next increment is developed by adding more features to the system.



We selected the Incremental Model because our project is large and consists of multiple modules and features. Developing the complete system at once would make the project difficult to manage. By using the Incremental Model, we can divide the system into smaller manageable parts and develop them step by step.

This model is suitable for our project because each increment can be developed and tested separately. It helps reduce complexity and allows the team to focus on one module at a time. If any problem is found in a module, it can be fixed before moving to the next increment.

Although some tasks in the Gantt chart are performed in parallel by different team members, the overall development process still follows the Incremental Model because the system is developed module by module in several increments.

Therefore, the Incremental Model is the most suitable process model for our Agricultural Equipment Rental and Management System

2. Timeline Chart / Gantt Chart

COCOMO Estimation (Organic Mode)

Given:

- SLOC = 17.8 KLOC
- Project Type = Organic
- Team Members = 5

Effort (PM):
PM = 2.4 * (KLOC)^{1.05}

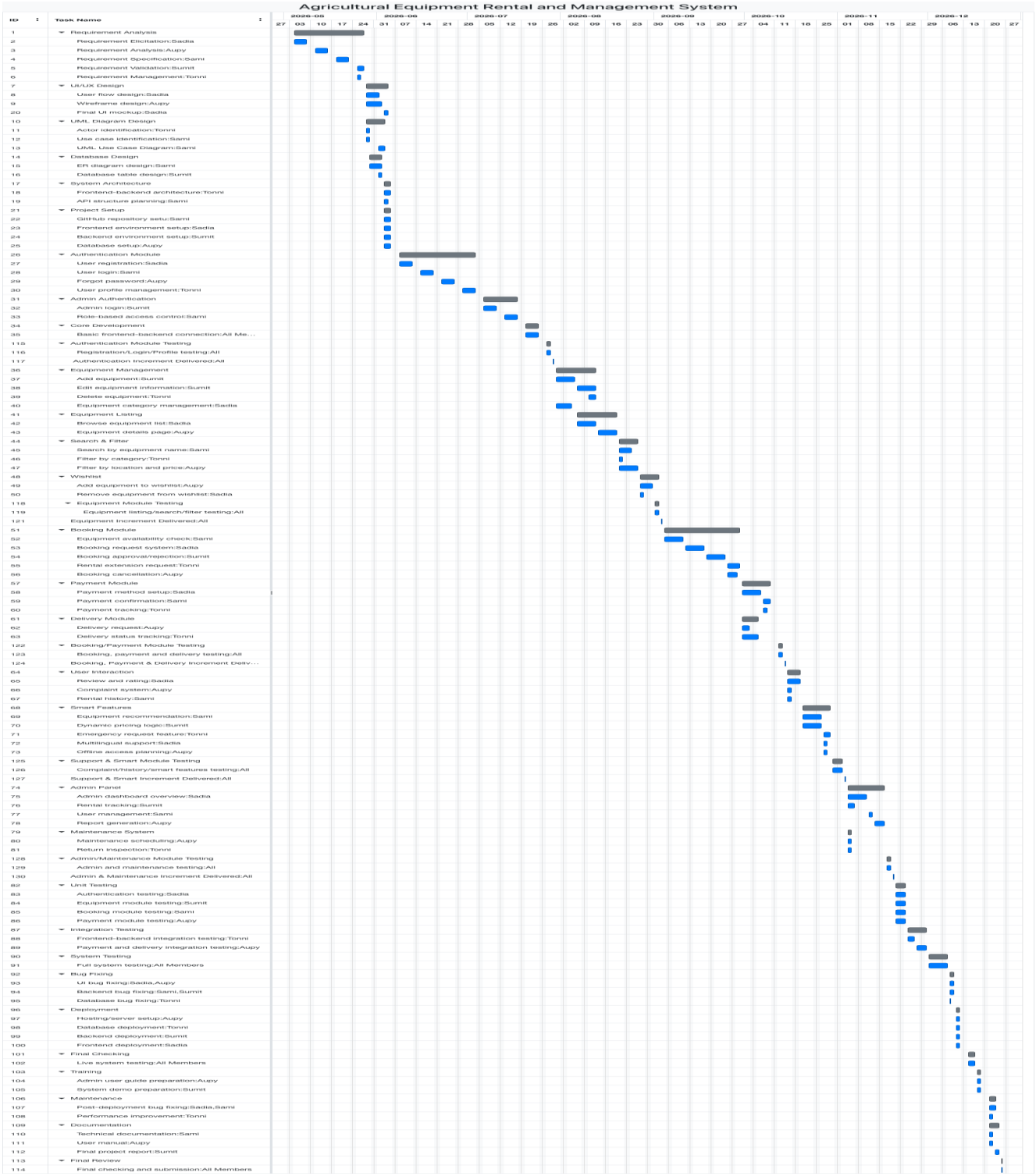
PM = 2.4 * (17.8)^{1.05}

PM=49.35 person-months

Development Time (DM):
DM = 2.5* (PM)^{0.38}

DM = 2.5 * (49.35)^{0.38}

DM =10.99 = 11 Months



Gantt chart link: https://www.onlinegantt.com/#/shared/01b22a57-6a80-4b83-8439-ac3fe8055bfa

3. REQUIREMENTS DOCUMENT (PRD)

3.1 Functional Requirements

Template Name	Agricultural Equipment Rental and Management System
Goals & Objective	The goal of this system is to provide a secure, efficient, and user-friendly platform for farmers, employees, and administrators to manage agricultural equipment rentals. From this system, users can: • Browse, rent equipment, check prices, and request rental extensions. • Submit emergency equipment requests, provide feedback, write reviews, and cancel rentals. • Access the system in multiple languages. • Make payments, manage profile information, and delete accounts. • Submit complaints regarding equipment or services.
Client Information	Internal University Project
Product Type	Web Application
Domain	Web2
Access Layers	Admin, Employee, Farmer
Design Required	Yes

Phase	Item No	Specification (Role)	Screen Definition	User Story	Acceptance Criteria
Feature: 1 – User Authentication		Auth Feature			
Phase 1	1	[Farmer, Admin, Employee] User Registration	Email & Password, Name, Phone Number field, Signup Button, Third Party SSO	As a user, I want to create an account so I can access the platform.	-The system should validate the email field for emptiness, correct format, and uniqueness, and display appropriate messages ("Email is required", "Invalid email address", "Email already used"). -The system should validate the password to ensure it is not empty and meets the minimum length requirement (at least 8 characters), otherwise display ("Password is required", "Invalid password"). -The system should ensure the re-password field matches the password and display ("Please confirm your password",

					"Password does not match") if validation fails. -The system should ensure a user type is selected before allowing signup, otherwise display ("Please select a user type"). -The system should prevent signup if any validation fails and displays the corresponding error message(s). -The system should store user credentials with an inactive status and generate a unique email verification link upon successful signup. -The system should send the verification link to the user's email and display ("Please verify your email") after signup. -The system should validate the verification link (with expiration) and activate the account if valid, otherwise display ("Invalid or expired verification link"). -The system should restrict login until email verification is completed and display ("Please verify your email before logging in") and allow access to the dashboard after successful verification. -The system should allow users to resend the verification email and enforce an expiration time for the verification link (e.g., 24 hours).
Phase 1	2	[Farmer, Admin,Employee] Login	Email/Username & Password, Third Party SSO Login	As a user, I want to log in securely.	- The system should validate that the email and password fields are not empty and display ("Email is required", "Password is required"). - The system should verify that the entered email exists and the password matches the registered credentials, otherwise display ("Email or password does not match!"). - The system should track failed login attempts and temporarily lock the account after a defined number of unsuccessful attempts, displaying ("Too many failed attempts. Please try again later"). - The system should ensure that only verified email accounts can log in and display ("Please verify your email before logging in") if not verified. - The system should securely store passwords using hashing mechanisms. - The system should allow login and redirect the user to the

					dashboard when credentials are correct and the account is active. - The system should notify the user in case of suspicious login activity.
Phase 1	3	[Farmer, Admin, Employee] Forgot Password	Email address, Verification Code, Confirm Button	As a user, I want to reset my password.	- The system should validate that the email field is not empty and exists in the database, otherwise display "Email is required" or "No email found!". - The system should send a password reset link to the registered email address and display "Password reset link sent!". - The system should validate the reset link when the user clicks it and display "Invalid or expired reset link" if the link is not valid. - The system should allow the user to create a new password only after successful reset link validation. - The system should ensure the new password is at least 8 characters long and includes at least one special character, otherwise display "Password must be at least 8 characters long and include a special character". - The system should ensure the re-type password field is not empty and matches the new password, otherwise display "Please re-type your password" or "Password does not match". - The system should update the password and display "Password changed" if all validations are successful.
Phase 1	4	[Farmer, Admin,Employee] Profile Management	User Name, Profile picture, Phone number, Email, Address, Update Button, Edit Button	As a user, I want to update my information.	- The system should allow the user to view and update basic profile information (name, phone number, address, profile picture) when the profile page is opened or the Edit option is selected; if the data fails to load, an error message should be shown: "Failed to load profile information. Please try again." - The system should provide clear “Edit” and “Update” options with accessible buttons and readable text; if the interface fails to respond, an error message should be displayed: "Unable to access profile editing options. Please try again." - The system should save the updated profile information when the user clicks the Update button; if the input data is invalid or the saving process fails, an error message should be shown: "Failed to update profile. Please ensure all fields are valid and try again."

					- The system should immediately display the updated information on the profile page after successful saving; if the update is not reflected, the system should refresh the data or show an error message: "Failed to refresh profile information." - The system should display a confirmation message after a successful update: "Profile updated successfully."
Feature: 2 – Product Management					
Phase 1	5	[Employee] Add Equipment	Equipment Image, Name, Price, Description, Quantity, Add Button	As an employee, I want to add new equipment.	- The system should require all mandatory fields (e.g., name, price, quantity, image) to be filled before submission and display appropriate error messages if any field is missing. - The system should ensure that an equipment image is uploaded and display an error message if not provided. - The system should validate that the rental price is a valid numeric value and display an error if the input is invalid. - The system should ensure that the quantity is a positive number and display an error if the value is zero or negative. - The system should display the rental price clearly on the booking page so users can review it before confirming the rental. - The system should confirm the rental details, including the price, and display them on the rental confirmation page after the user clicks "Rent". - The system should display a success message ("Equipment added successfully") after successful submission.
Phase 1	6	[Farmer] Browse Equipment	Equipment List with Name, Price, Image, Availability, Search & Filter Options	As a farmer, I want to search and view available equipment.	- The system should display the equipment list with clear details such as name, image, rental price, and availability. - The system should display a message ("No equipment available") if no equipment exists in the system. - The system should allow users to search and filter equipment based on relevant attributes such as type, price, and location. - The system should update the equipment list dynamically when a search or filter is applied. - The system should display a message ("No equipment found") if no equipment matches the selected search or filter criteria.

					- The system should clearly indicate the availability status of each equipment item using simple labels such as "Available" or "Not Available". - The system should allow users to navigate to the equipment details page by clicking on an equipment item or a "View Details" option.
Phase 1	7	[Admin] Managing and organizing equipment	Equipment List, Add Button, Update Button, Remove Button	As an admin, I want to manage equipment details.	- The system should display each equipment's name, price, image, and availability upon opening the equipment list. If the data is missing or loading failure, an error message should be shown: "Failed to load equipment list. Please try again." - The system should open a form to input equipment details when the Add Equipment button is clicked, if any required field is empty or invalid, an error message should be displayed: "All fields are required and must be valid." - The system should save the new equipment and display it in the list after valid equipment details are submitted; if the saving process fails, an error message should be shown: "Failed to save equipment. Please try again." - The system should allow editing of equipment details when the Update option is selected, if the updated data is invalid or the update process fails, an error message should be displayed: "Failed to update equipment. Please ensure all fields are valid and try again." - The system should delete the selected equipment from the list when the Remove button is clicked and ensure it no longer appears in the equipment list after successful deletion, if the deletion fails, an error message should be shown: "Failed to delete equipment." and the system should refresh the list if it does not update properly.
Phase 1	8	[Admin] Manage Categories	Add/Edit/Delete Equipment Categories	As an admin, I want to manage the equipment categories.	- The system should display all equipment categories upon opening the category list; if the data is missing or loading fails, an error message should be shown: "Failed to load categories. Please try again." - The system should open a form to input a new category when the Add Category button is clicked; if the category name is empty or already exists, an error message should be displayed: "Category name cannot be empty or duplicated." - The system should save the new category and display it in the list

					after valid input is submitted; if the saving process fails, an error message should be shown: "Failed to save category. Please try again." - The system should allow editing of category details when the Update option is selected; if the updated name is empty, duplicated, or the update process fails, an error message should be displayed: "Failed to update category. Please ensure the name is valid and try again." - The system should delete the selected category from the list when the Remove button is clicked and ensure it no longer appears in the category list after successful deletion; if the deletion fails, an error message should be shown: "Failed to delete category." and the system should refresh the list if it does not update properly. - The system should display a confirmation message after each successful action (add, update, delete): "Action completed successfully."
Feature 3 – Booking & Rental					
Phase 1	9	[Farmer] Booking equipment for rental	Equipment Details, Rental Duration, Booking Details, Confirm Button	As a farmer, I want to book equipment for a specific duration.	- The system should confirm the booking upon selecting equipment and rental period, if the booking process fails, the system should allow the user to attempt the process again and verify all required selections with an error message: "Booking could not be completed. Please try again." - The system should clearly display the total price upon selecting a rental period, if the price cannot be calculated, the system should recalculate based on valid input or require the user to reselect the rental period with an error message: "Unable to calculate price. Please try again." - The system should send a confirmation email to the farmer including rental details after successful booking, if the email fails to send, the system should attempt to send the email again or notify the user to verify their email address with an error message: "Booking confirmed, but email could not be sent." - The system should save the booking details and make them visible in the farmer's rental history after successful booking, if saving fails, the system should attempt the saving process again or notify the user to perform the action again with an error

					message: "Failed to save booking details."
Phase 1	10	[Admin]Managing and tracking rentals	List of Booked Rentals, Status Button,	As an admin, I want to manage all equipment rentals.	- The system should display a list of rentals to the admin including equipment name, rental status, and user details, if the data fails to load, the system should reload the rental data or allow the admin to refresh the list and display an error message: "Failed to load rental list. Please try again." - The system should allow the admin to mark rentals as delivered or returned, if the update process fails, the system should keep the previous status unchanged, allow the admin to submit the update again, and display an error message: "Failed to update rental status." - The system should display the rental status as Pending, Delivered, or Returned for each rental; if the status is not updated correctly, the system should reload the latest status from the database, ensure synchronization, and display an error message: "Unable to display updated status."
Phase 1	11	[Farmer] Rental Extension	Extension Button	As a farmer, I want to extend my equipment rental.	- The system should allow the user to request a rental extension before the expiry date; if the request is made after expiry or fails to process, an error message should be shown: "Extension request cannot be processed. Please try again before the rental expires." - The system should check equipment availability for the requested extended period after the extension request is submitted; if the equipment is not available, an error message should be displayed: "Equipment is not available for the selected extended period." - The system should display the updated rental duration and total price clearly after a valid extension request is made; if the calculation fails or data is missing, an error message should be shown: "Failed to calculate updated rental details. Please try again." - The system should confirm the extension request after successful approval or payment; if the approval or payment process fails, an error message should be displayed: "Extension request failed. Please complete the payment or try again." - The system should save the updated rental details after a successful extension; if saving fails, an error message should be

					shown: "Failed to update rental details. Please try again." - The system should send a confirmation notification to the user after a successful extension; if the notification fails to send, an error message should be displayed: "Extension successful but failed to send confirmation notification."
Feature 4 – Payment & Delivery Management					
Phase 1	12	[Farmer] Making payments for rentals	Payment Methods (Card, Mobile Wallet), Apply Coupon, Confirm Payment Button	As a farmer, I want to securely pay for my rental.	- The system should provide multiple payment options including credit cards and mobile wallets if a selected payment method is unavailable or fails to load, the system should display an error message: "Selected payment method is not available." and allow the user to choose an alternative payment option. - The system should confirm the payment after a successful transaction if the transaction fails, the system should cancel the process, ensure no amount is deducted, and display an error message: "Payment failed. Please try again." - The system should send an email receipt to the user upon successful payment including transaction details, if the email fails to send the system should store the receipt and notify the user with an error message: "Payment successful, but receipt email could not be sent." - The system should allow users to apply a valid coupon to reduce the total price if the coupon is invalid or expired, the system should display an error message: "Invalid or expired coupon code." and maintain the original price.
Phase 1	13	[Admin] Tracking and managing payment activities	Payment Status, Invoice Generation, Payment History	As an admin, I want to track and manage payments.	- The system should allow the admin to view the payment status for each rental, if the payment data is unavailable or fails to load, the system should display an error message: "Unable to load payment status." and allow the admin to refresh or reload the data. - The system should allow the admin to generate invoices for rentals including all relevant payment details, if the invoice generation fails, the system should display an error message: "Failed to generate invoice." and allow the admin to attempt the action again. - The system should allow the admin to track payment history for each rental including transaction details, if the data cannot be retrieved, the system should

					display an error message: "Failed to load payment history." and allow the admin to reload the information. - The system should notify the admin if a payment fails including relevant rental and user details if the notification cannot be delivered, the system should log the failure and display an error message: "Payment failure notification could not be delivered." - The system should allow the admin to export payment history and invoices in a downloadable format. If the export process fails, the system should display an error message: "Failed to export payment data." and allow the admin to attempt the export again.
Phase 1	14	[Admin] Managing equipment delivery process	Order List, Delivery Status, Delivery Team Contact	As an admin, I want to manage delivery and returns.	- The system should display the delivery status for all equipment including statuses such as In Progress and Delivered ,if the delivery data is unavailable or fails to load, the system should display an error message: "Unable to load delivery status." and allow the admin to refresh the data. - The system should allow the admin to track the progress of the delivery team for each rental ,if the tracking data cannot be retrieved, the system should display an error message: "Failed to load delivery progress." and allow the admin to reload the information. - The system should provide the admin with access to delivery team contact details including phone and email if the contact details are unavailable, the system should display an error message: "Unable to load contact details." and allow the admin to refresh the data. - The system should allow the admin to update the delivery status to In Progress or Delivered if the update process fails, the system should keep the previous status unchanged, display an error message: "Failed to update delivery status." and allow the admin to perform the action again. - The system should send a notification to the admin when a delivery is completed if the notification fails to send, the system should log the failure and display an error message: "Delivery notification could not be sent."
Feature: 5 – Smart Recommendation & Intelligent Services					

Phase 2	15	[Farmer, Admin] Smart Equipment Matching	Input Fields (Farm Size, Crop Type, Season), Suggestion List, Select Button	As a farmer, I want the system to suggest suitable equipment so I can save time and effort.	- If all inputs (Farm Size, Crop Type, Season) are valid → system displays relevant equipment based on predefined mapping rules (e.g., Rice + Harvest → Harvester) - If Farm Size is empty or invalid → system shows validation error "Farm size is required" and disables suggestions - If Crop Type is not found in database → system displays "No matching equipment found" - If Season is not selected → suggestion list remains inactive - If user changes any input → system dynamically refreshes suggestions in real time without page reload - If multiple equipment match → system ranks and displays top relevant options - If valid equipment is selected → user can proceed to booking/request action
Phase 2	16	[Admin] Seasonal Equipment Pricing Management	Price Display, Seasonal Indicator, Updated Price Field	As a system, I want to adjust equipment prices based on season so both farmers and owners benefit.	- If Season = Harvest → system increases price based on predefined rule (e.g., Base Price + 20%) - If Season = Off-season → system decreases price (e.g., Base Price – 10%) - If Base Price is missing or null → system shows error and prevents price calculation - If invalid or unknown season input → system ignores adjustment and displays base price - If admin updates pricing rules → system applies changes instantly across all relevant equipment - System ensures final price is never negative or zero - System logs all price changes for audit purposes
Phase 2	17	[Farmer, Admin] Emergency Equipment Request Management	Emergency Button, Location Detection, Equipment List	As a farmer, I want to request equipment urgently so I can handle time-critical tasks.	- If location access is enabled and equipment is available nearby → system identifies nearest equipment and assigns request automatically - If multiple equipment available → system selects the closest based on distance algorithm - If location access is disabled → system prompts user to enable GPS/location services - If no equipment available in nearby radius → system displays "No equipment available nearby" - If request is successfully sent → user receives instant confirmation message - System sends notification to admin/employee for approval or monitoring - If request fails due to network/system issue → system provides retry option

Phase 2	18	[Admin] Analytics Dashboard Management	Dashboard View (Revenue, Bookings, Usage Graphs), Filter Option	As an admin, I want to see analytics so I can make better decisions.	- If valid date range is provided → system displays revenue, bookings, and usage graphs accordingly - If equipment filter is applied → system shows filtered data specific to selected equipment - If invalid date range (end date earlier than start date) → system shows validation error - If no data available for selected filters → system displays "No data available" message - Dashboard must load within acceptable time without errors - Graphs must accurately represent stored data - System allows real-time or periodic data refresh
Phase 2	19	[Admin, Employee] Equipment Health Monitoring and Maintenance Management	Equipment Status, Usage History, Maintenance Alert	As a system, I want to monitor equipment health so that breakdowns can be prevented.	- If equipment usage exceeds predefined threshold → system automatically generates maintenance alert - If equipment is recently serviced → system sets status as "Good" - If maintenance is due → system updates status to "Needs Maintenance" - If usage data is missing or incomplete → system displays warning and flags record - System stores all maintenance history and usage logs persistently - Alerts must be visible to admin/ employee in dashboard - System prevents booking of equipment marked as "Needs Maintenance" (optional rule if applied)
Phase 2	20	[Farmer, Admin] Multilingual and Offline Access Management	Language Selection (Bangla, English), Offline Indicator, Sync Button	As a farmer, I want to use the system in my preferred language and access basic features offline.	- If Bangla is selected → entire UI is translated to Bangla consistently - If English is selected → system switches back to English UI - If offline mode is active → user can access basic features (browse, request) with locally stored data - If internet connection is restored → system automatically synchronizes offline data with server - If synchronization fails → system shows error message and retry option - System displays sync status (pending/success/failed) - System should ensure no data loss during offline-to-online transition
Feature: 6 – Maintenance & Inspection System					
Phase 3	21	[Employee] Schedule Equipment Maintenance	Equipment List, Maintenance Date, Status (Scheduled/In Maintenance/Completed), Save Button	As an employee, I want to schedule maintenance so that	- The system should allow employees to assign maintenance dates. If maintenance scheduling fails or not completed, the system does not assign any date and

				equipment remains usable and safe.	displays an error message: "Maintenance scheduling failed." - The system should restrict booking for equipment under maintenance. If maintenance is not active or not scheduled, the system allows the equipment to be booked. - The system should display and updates maintenance status when maintenance is scheduled or in progress. If maintenance information is unavailable or not set, the system does not display any maintenance status. The system saves maintenance history after maintenance activities occur. If no maintenance activity is recorded or saving fails, the system does not store any history. - The system should display a success message after maintenance is successfully scheduled. If scheduling fails or is incomplete, the system displays an error message: "Maintenance scheduling failed."
Phase 3	22	[Employee] Inspect Returned Equipment	Return List, Condition Options (Good/Damaged/Needs Service), Comment Box, Submit Button	As an employee, I want to inspect returned equipment so I can determine its condition.	- The system should allow employees to select the condition after return. If the condition is not selected or the process fails, the system does not record the condition and displays an error message: "Condition selection failed." - The system should allow users to add additional notes in the comment field. If the comment field is not filled or fails to save, the system does not store any additional notes. - The system should mark damaged equipment as unavailable. If the equipment is not marked as damaged, the system keeps it available for use. - The system should store inspection data after the return process. If inspection data is not recorded or saving fails, the system does not store any inspection information. - The system should display a confirmation message after successful submission. If the process fails or is incomplete, the system displays an error message: "Confirmation failed."
Feature: 7 – User Interaction & Support					
Phase 3	23	[Farmer] Request Rental Cancellation	Booking List, Cancel Button, Reason Input, Submit Button	As a farmer, I want to cancel my booking so I can adjust my plans.	- The system should allow cancellation only before delivery. If the delivery is already completed or in progress, the system does not allow cancellation and displays an error message: "Cancellation not allowed after delivery."

					- The system should require the user to provide a reason for cancellation. If no reason is provided or submission fails, the system does not process the cancellation and displays an error message: "Cancellation reason required." - The system should update the booking status to "Cancelled" after a successful cancellation. If the cancellation process fails, the system does not update the booking status. - The system should send a cancellation notification to the admin after a successful cancellation. If notification delivery fails, the admin does not receive any alert. - The system should keep cancelled bookings in the history. If the saving process fails, the system does not store the cancelled booking in history.
Phase 3	24	[Farmer] Submit Equipment Review & Rating	Rating Stars (1–5), Review Text Box, Submit Button	As a farmer, I want to rate equipment so I can share my experience.	- The system should allow the user to select a rating between 1 to 5 stars. If no rating is selected or the process fails, the system does not submit the rating and displays an error message: "Rating selection required." - The system should allow the user to write a review in the text box. If the review field is empty or the submission fails, the system does not submit the review and displays an error message: "Review description cannot be empty." - The system should store the rating and review after successful submission. If the saving process fails, the system does not store the review and displays an error message: "Review submission failed." - The system should display a confirmation message after successful submission. If the process is completed, the system displays a message: "Review submission successful."
Phase 3	25	[User] Manage Complaint & Support Tickets	Complaint Form, Category Dropdown, Description Box, Submit Button	As a user, I want to report issues so they can be resolved quickly.	- The system should allow the user to select a complaint category. If no category is selected or the process fails, the system does not submit the complaint and displays an error message: "Complaint category selection failed." - The system should require the description field to be filled before submission. If the description is empty or the submission fails, the system does not submit the complaint and displays an error message: "Description cannot be empty." - The system should generate a unique ticket ID after successful complaint submission. If ticket

					generation fails, the system does not create the complaint record and displays an error message: "Ticket ID generation failed." - The system should allow the admin to update the complaint status. If the update process fails or is incomplete, the system does not change the complaint status and displays an error message: "Complaint status update failed." - The system should display complaint status updates to the user. If the status is not updated or cannot be retrieved, the system does not show any new complaint status information.
Phase 3	26	[Farmer] View & Download Rental History Report	Rental List, Filter Options, Download Button (PDF)	As a farmer, I want to view my rental history so I can track my activities.	- The system should display all rentals. If rental data is unavailable or cannot be retrieved, the system does not display the rental list and shows an error message: "Unable to load rental data." - The system should apply filters properly based on date and status. If the filter process fails or invalid filter criteria are used, the system does not display the filtered results and shows an error message: "Filter operation failed." - The system should generate a report when the user selects the download option. If report generation fails, the system does not download the report and shows an error message: "Report generation failed." - The system should include equipment name, duration, and price in the report. If any required report data is missing or cannot be retrieved, the system does not generate a complete report and shows an error message: "Incomplete report data." - The system should ensure that the report data matches the system records. If the report data does not match the stored records, the system marks the report as invalid and shows an error message: "Report data mismatch detected."
Phase 3	27	[Employee] Verify Farmer Account	User Verification List, Farmer Details, Document Upload/View, Approve Button, Reject Button	As an employee, I want to verify farmer accounts so that only valid users can rent equipment.	- The system should display a list of newly registered farmers waiting for verification. If no pending farmer account is found or the data cannot be retrieved, the system does not display the list and shows an error message: "Unable to load verification requests." - The system should allow the employee to view farmer details such as name, phone number, address, and submitted document. If the farmer details or document cannot be loaded, the system does not display the details and shows an error message: "Unable to load farmer details."

					- The system should allow the employee to approve a farmer account after checking the submitted information. If the approval process fails, the system does not change the account status and shows an error message: "User verification failed. Please try again." - The system should allow the employee to reject a farmer account if the submitted information is invalid or incomplete. If the rejection process fails, the system does not change the account status and shows an error message: "User verification failed. Please try again." - The system should allow approved farmers to book equipment. If the farmer account is pending or rejected, the system does not allow equipment booking and shows an error message: "Account verification required before booking." - The system should show the verification status to the farmer after approval or rejection. If the status cannot be retrieved, the system does not display the verification result and shows an error message: "Unable to load verification status." - The system should send a notification to the farmer after approval or rejection. If the notification fails, the system saves the verification result but shows an error message: "Verification completed, but notification could not be sent."
Feature 8 – System Administration					
Phase 1	28	[Admin] Generate Reports	Sales, Payment, and Rental Reports	As an admin, I want to generate system reports for analytics	- The system should allow the admin to generate reports (sales, payment, rental) when the report option is selected; if the report generation fails, an error message should be shown: "Failed to generate report. Please try again." - The system should display accurate and up-to-date data in the generated reports; if the data is missing or incorrect, an error message should be displayed: "Report data is unavailable or inaccurate. Please refresh and try again." - The system should allow the admin to filter reports by selecting a date range; if the selected date range is invalid or no data is found, an error message should be shown: "Invalid date range or no data available for the selected period."

					- The system should allow downloading reports in supported formats (PDF/CSV) after successful generation; if the download fails, an error message should be displayed: "Failed to download report. Please try again." - The system should ensure that generated reports match the system records; if inconsistencies are detected, an error message should be shown: "Report data mismatch detected. Please verify system records." - The system should display a confirmation message after successful report generation: "Report generated successfully."
Feature 9 – Account & Security Management					
Phase 4	29	[Farmer] Account Deletion	Account Deletion Option	As a farmer, I want to delete my account.	- The system should allow the user to request account deletion when the delete account option is selected; if the request fails, an error message should be shown: "Failed to process account deletion request. Please try again." - The system should ask for confirmation before deleting the account; if the user cancels the action, the account should remain unchanged. - The system should allow only authenticated users to delete their account; if an unauthenticated user attempts this action, an error message should be displayed: "You must be logged in to delete your account." - The system should remove or mark all user data as inactive after successful account deletion; if the process fails, an error message should be shown: "Failed to delete account data. Please try again." - The system should display a confirmation message after successful account deletion: "Your account has been deleted successfully." - The system should prevent login attempts using deleted accounts; if a deleted account is used to log in, an error message should be displayed: "This account no longer exists or has been deactivated."
Feature 10 – User Personalization					
Phase 4	30	[Farmer] Wishlist / Save Equipment	Save Button, Wishlist Page, Remove Option	As a farmer, I want to save equipment so I can easily access them later.	- The system should allow the user to save equipment when the Save or Favorite button is clicked; if the process fails, an error message should be shown: "Failed to save equipment. Please try again."

					- The system should display the saved equipment in a dedicated saved list or favorites section; if the data fails to load, an error message should be displayed: "Failed to load saved equipment. Please try again." - The system should prevent duplicate saving of the same equipment; if the user tries to save an already saved item, an error message should be shown: "Equipment is already saved." - The system should allow the user to remove equipment from the saved list; if the removal process fails, an error message should be displayed: "Failed to remove saved equipment. Please try again." - The system should ensure saved equipment data is updated and reflects current availability and details; if the update fails, an error message should be shown: "Failed to update saved equipment details." - The system should display a confirmation message after successfully saving or removing equipment: "Action completed successfully."

3.2 Non-Functional Requirements

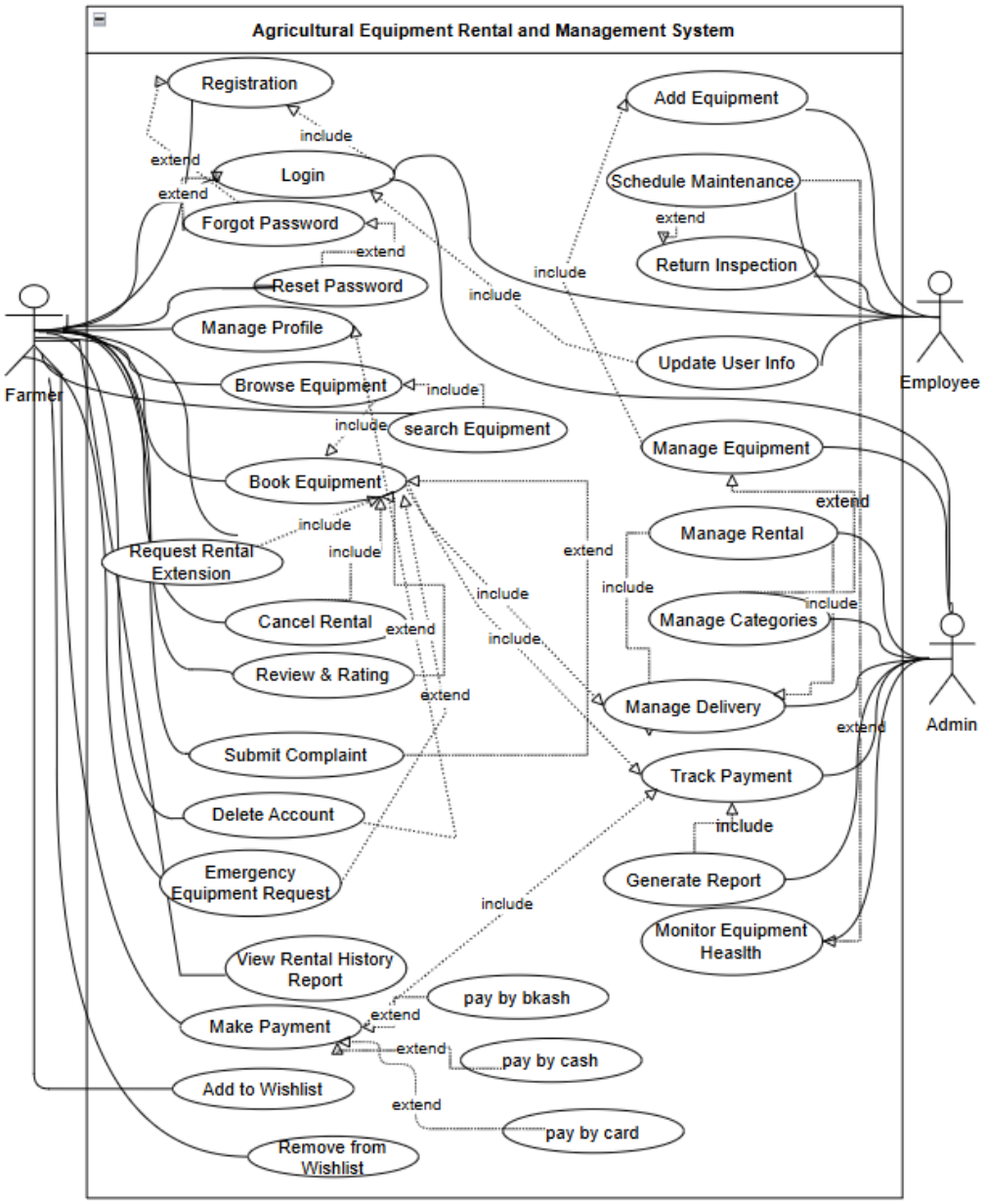
Category	Item No	Specification	User Story	Acceptance Criteria
Performance	31	Fast Homepage Loading	As a user, I want the homepage to load quickly so that I can access information without waiting.	The homepage **should** load in 3 seconds on 4G
Performance	32	Fast API Responses	As a user, I want API calls to be fast so that I can get data without delays.	90% of API calls should finish in under 500ms.
Reliability	33	System Availability	As a user, I want the system to always be available so that I can use it anytime without interruption.	The system must maintain 99.5% uptime every month.
Reliability	34	Quick Recovery	As a user, I want the system to recover quickly from failures so that I can continue using it without long interruptions.	The system should recover from crashes within 2 minutes.
Scalability	35	Handle Many Users	As a user, I want the system to handle many users at once so that performance remains stable during high traffic.	The system must handle 50,000 concurrent users.
Security	36	Safe Passwords	As a user, I want the system to encrypt all passwords so that my personal data remains secure.	stored in hashed format using modern encryption algorithms.
Security	37	OTP Timeout	As a user, I want the system to implement OTP verification for password reset so that my account remains secure.	OTP should expire after 120 seconds.
Security	38	Fraud Detection	As a user, I want the system to detect suspicious activity so that my account is protected from unauthorized access.	Alert the admin if suspicious activity is detected, based on user patterns.

Usability	39	Mobile Friendly	As a user, I want the system to be fully responsive on mobile devices so that I can use it easily on any screen size.	All pages should work seamlessly on mobile devices.
Usability	40	User-Friendly Navigation	As a user, I want a simple and easy-to-understand navigation system so that I can quickly find and use features without confusion.	The system should have a simple and easy to understand interface for everyone.
Maintainability	**41**	Easy System Maintenance	As a developer, I want the system to be easy to maintain and update so that I can fix bugs and add new features without affecting existing functionalities.	The system should allow developers to fix bugs and update features without affecting existing functionalities.
Compatibility	**42**	Cross-Platform Support	As a user, I want the system to work on different platforms so that I can access it from any device or operating system.	The system should run properly on Windows, macOS, and major web browsers without errors.
Security	**43**	Two-Factor Authentication	As a user, I want two-factor authentication for login so that my account remains secure from unauthorized access.	The system should offer 2FA for user logins.
Availability	**44**	Offline Handling	As a user, I want the system to handle temporary network issues so that I do not lose data during connectivity problems.	The system should show proper error messages and prevent data loss during network failure.
Performance	**45**	Fast Maintenance Update	As an admin, I want maintenance updates to be processed quickly so that changes are reflected without delay.	Maintenance updates should be saved and reflected within 2 seconds
Reliability	**46**	Accurate Inspection Data	As an employee, I want inspection data to be accurate and reliable so that I can make correct decisions about equipment condition.	Inspection records should save without data loss and remain consistent in the system.
Security	**47**	Secure Complaint Handling	As a user, I want my complaint data to be secure so that my personal information remains protected.	Only authorized users (Admin) should access complaint details
Usability	**48**	Easy Review Submission	As a user, I want to easily submit reviews so that I can quickly share my feedback without confusion.	Users should be able to submit a rating and review within 3 steps
Maintainability	**49**	Report Generation Stability	As an admin, I want reports to generate reliably so that I can analyse rental data without errors.	Reports should generate without errors and match with stored data
Usability	**50**	Clear Error Messages	As a user, I want clear error messages so that I can understand and fix issues easily.	All error messages should simple, specific, and guide users on how to fix the issue

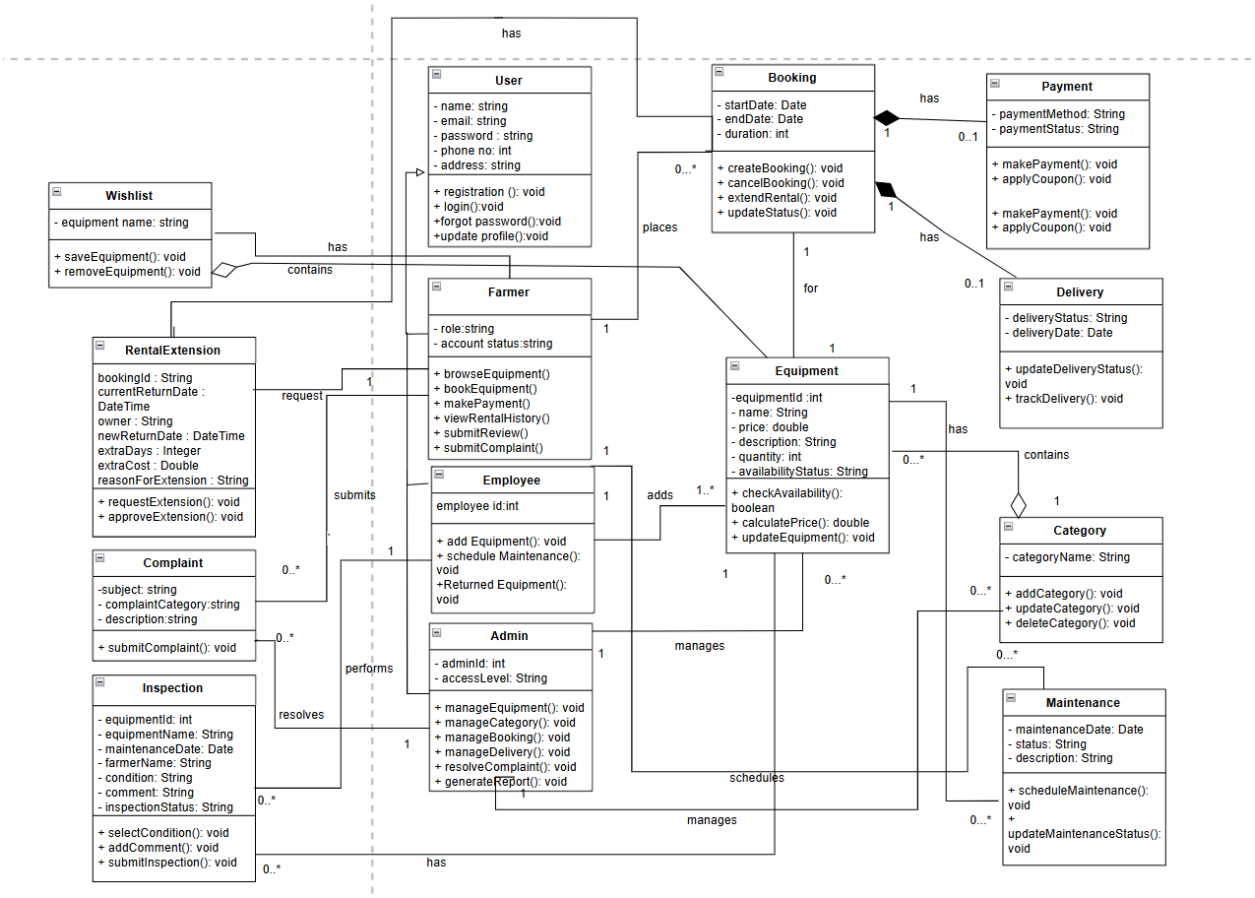
4. SOFTWARE DESIGN

4.1 System Design:

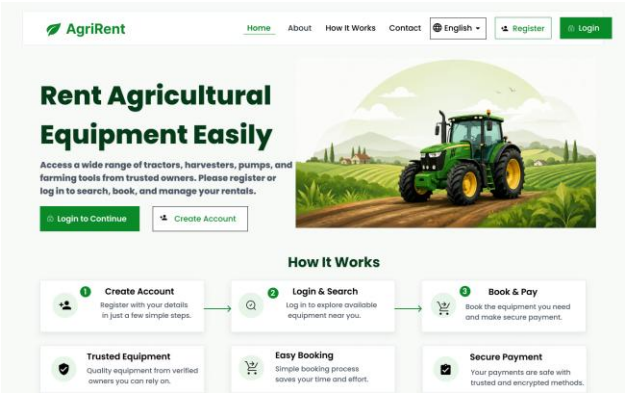
Use Case Diagram



Class Diagram



1.1 UI / Wireframe Design using Figma



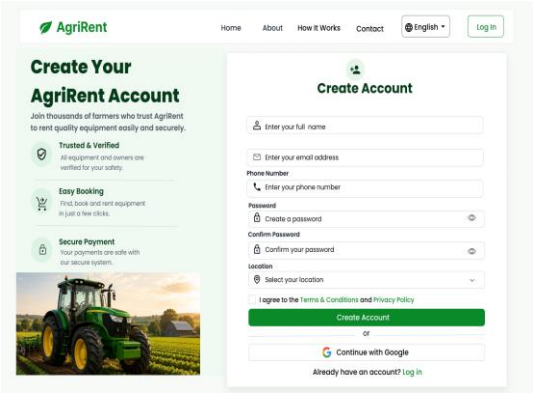
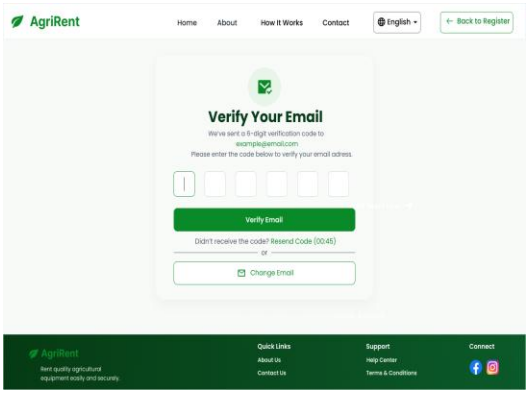


Fig 1: Home page

Fig 2: Registration page



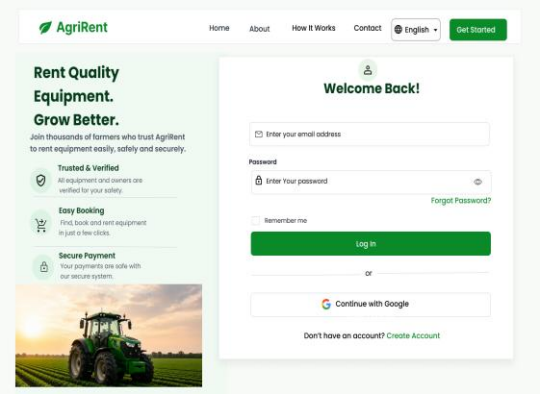
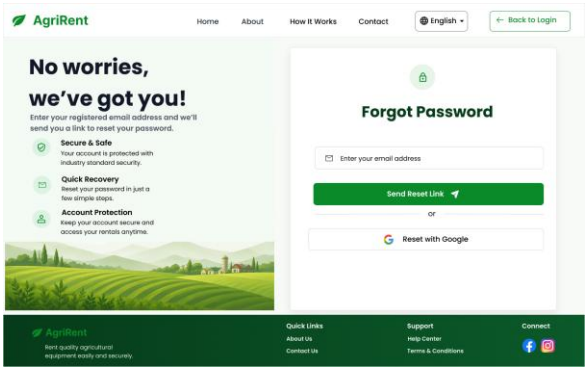


Fig 3: Email verification page

Fig 4: Login page



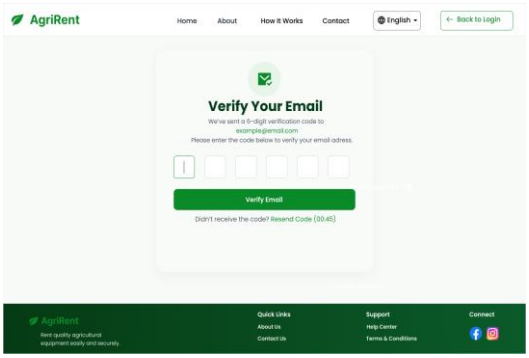
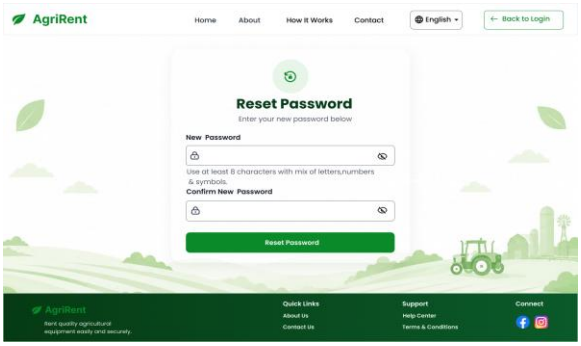


Fig 5: Forget password page

Fig 6: verification reset page



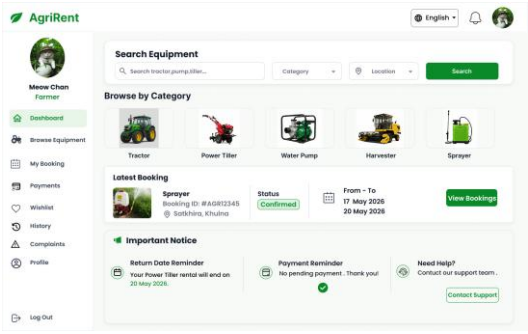


Fig 7: Reset password page

Fig 8: Farmer dashboard

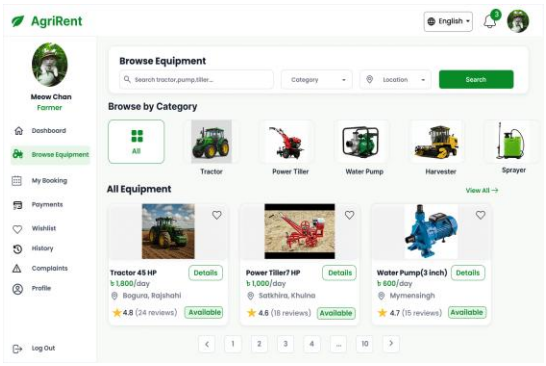


Fig 9: Browse Equipment page

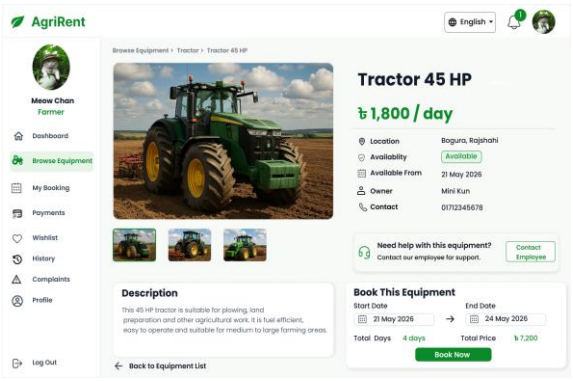


Fig 10: Equipment details page

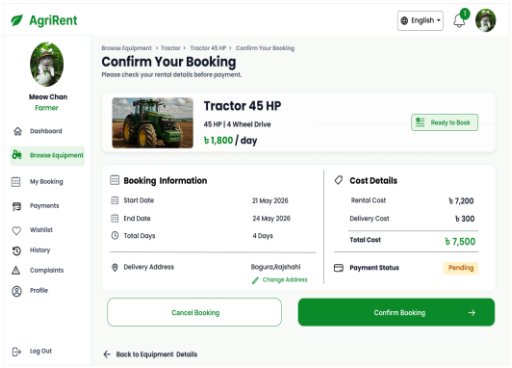


Fig 11: Confirm Booking page

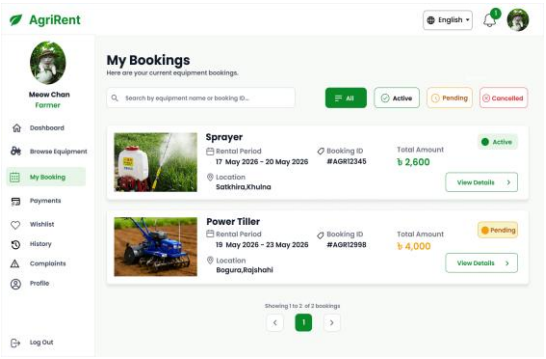


Fig 12: My booking page

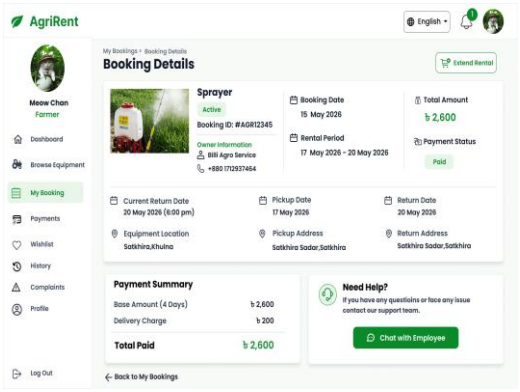


Fig 13: Booking details page

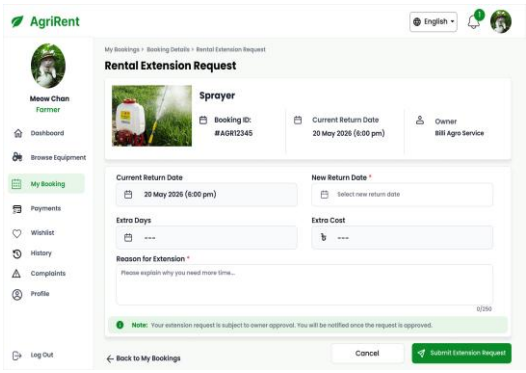


Fig 14: Rental extension page

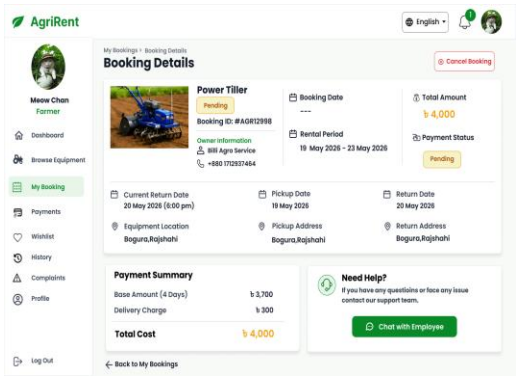


Fig 15: Booking details page for pending

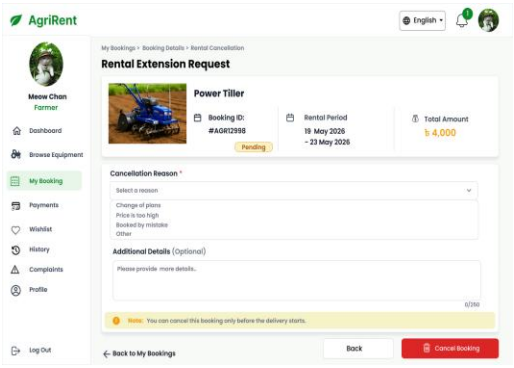


Fig 16: Rental Cancellation page

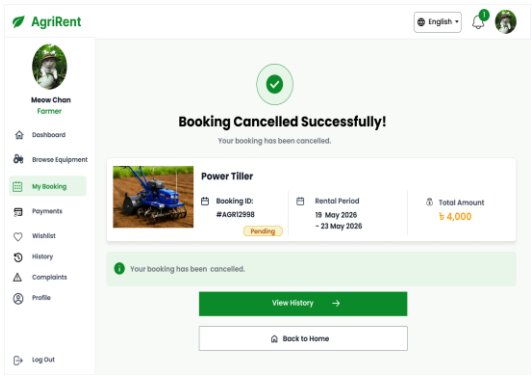


Fig 17: Rental cancel successfull page

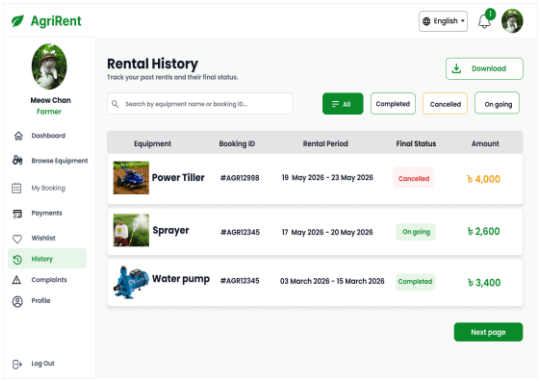


Fig 18: Rental History page

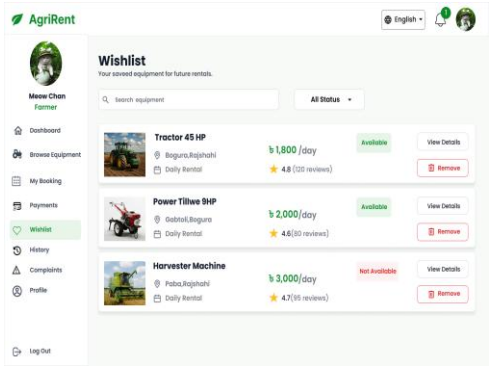


Fig 19: Wishlist page

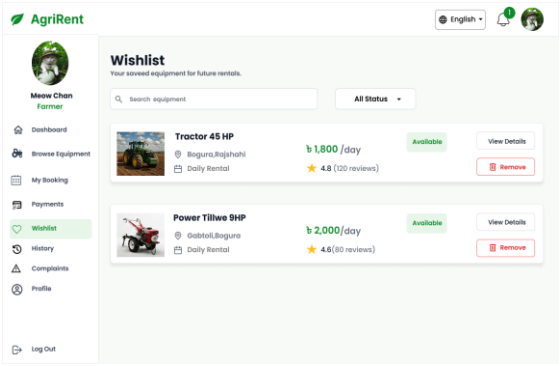


Fig 20: Wishlist remove page

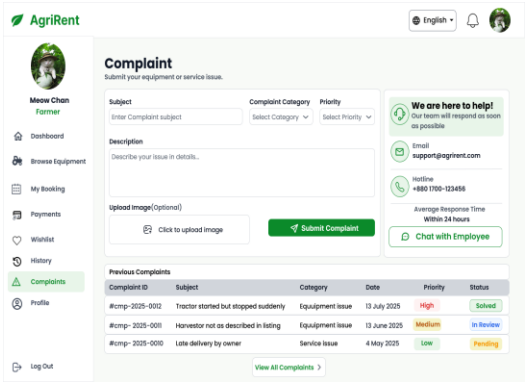


Fig 21: Complaint page

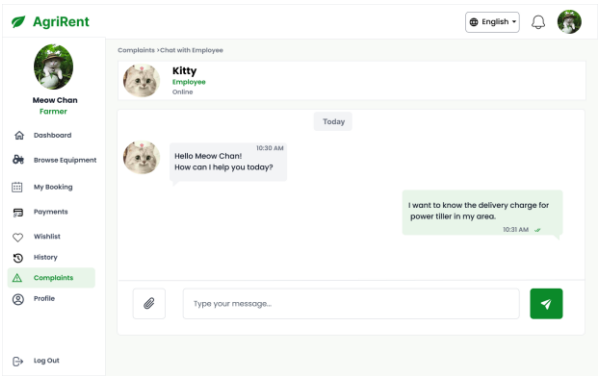


Fig 22: Chat with employee page

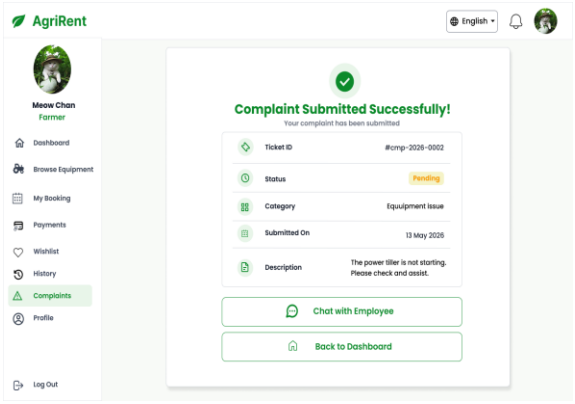


Fig 23: Complaint successfullpage

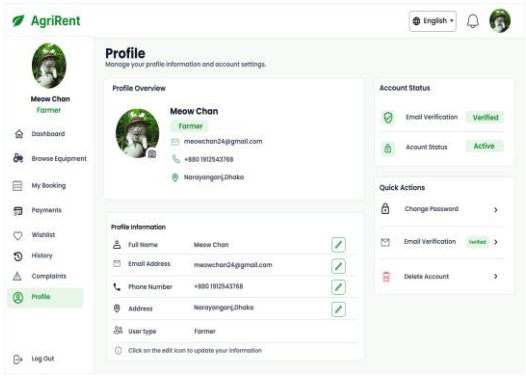


Fig 24: Profile edit page

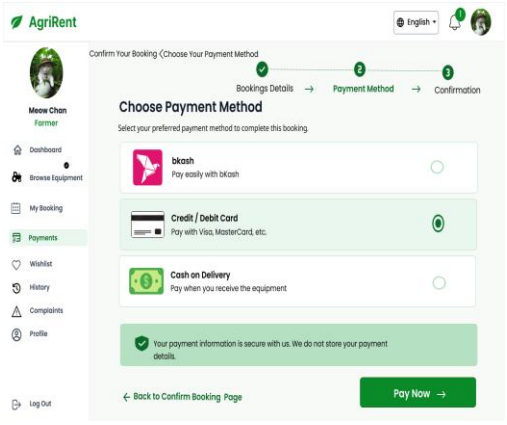


Fig 25:Payment method card page

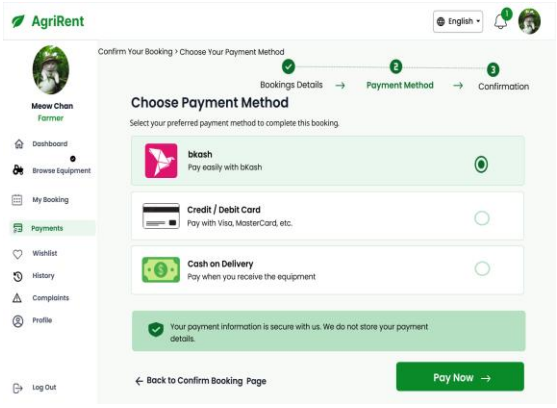


Fig 26: Payment method page

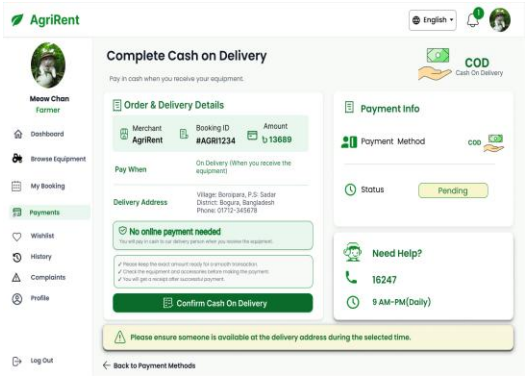


Fig 27:Cash on delivery page

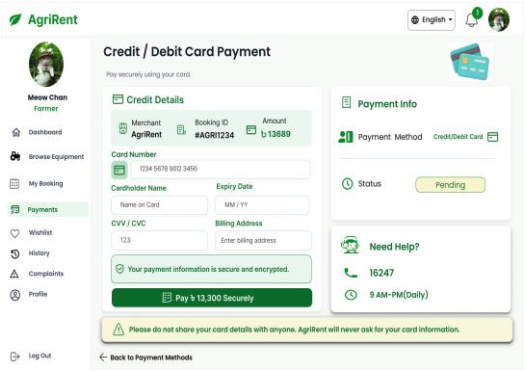


Fig 28: Debit card method page

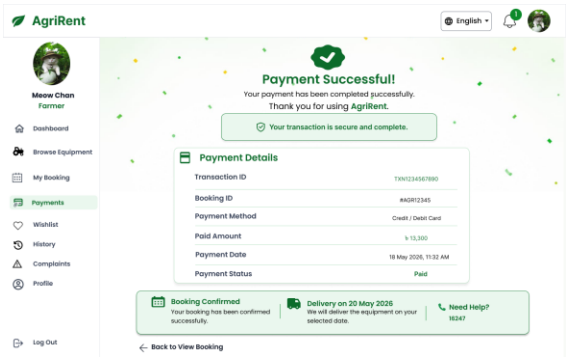


Fig 29: debit card Payment successful page

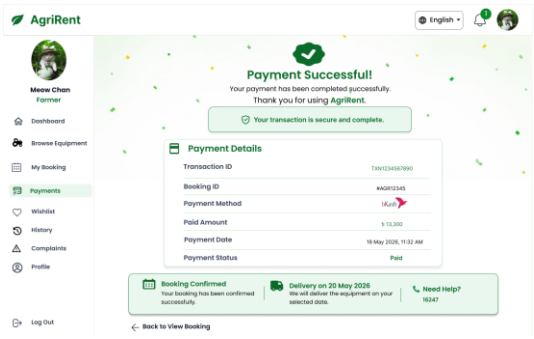


Fig 30: Bkash payment Successful page

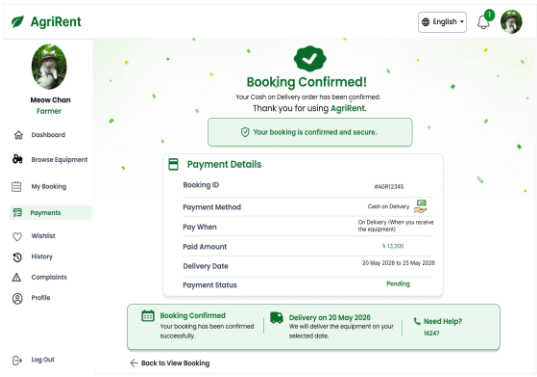


Fig 31: cash on Payment successful page

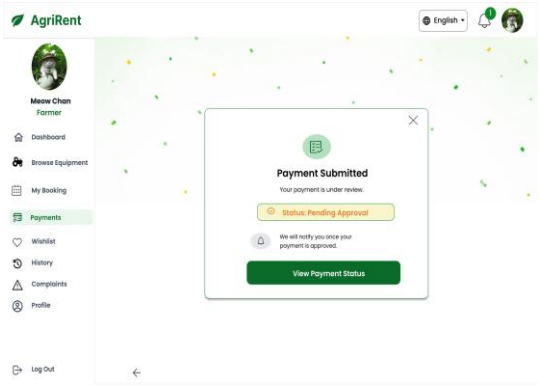


Fig 32: Pop up page

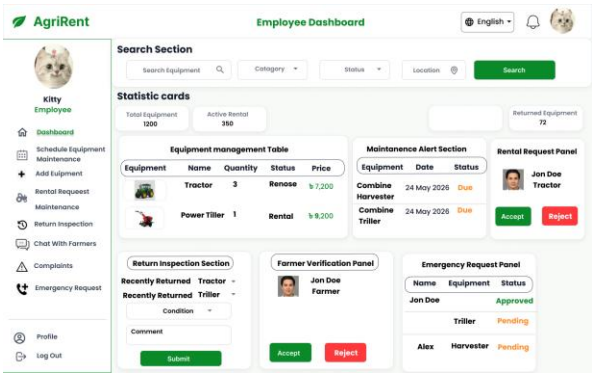


Fig 33: Employee dashboard page

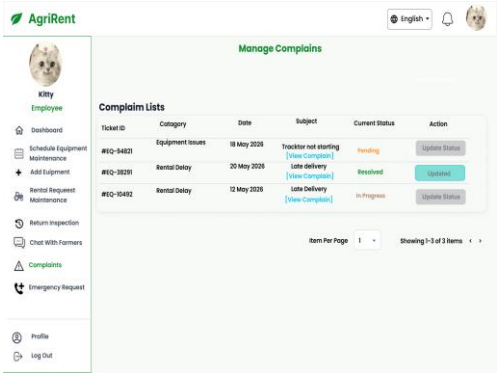


Fig 34:Manage complaints page

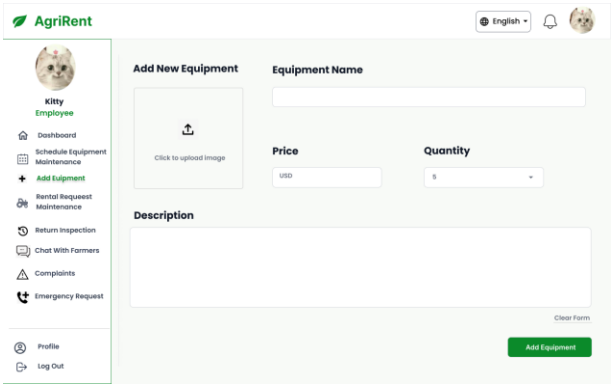


Fig 35: Add Equipment page

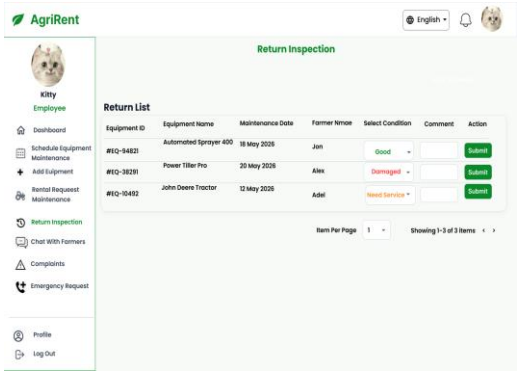


Fig 36:Return inception page

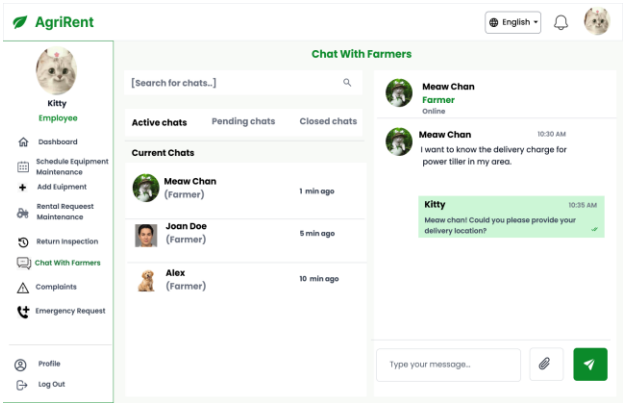


Fig 37: Chat with farmer page

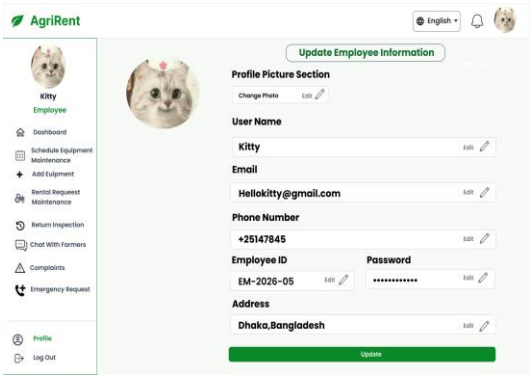


Fig 38:Update employee info page

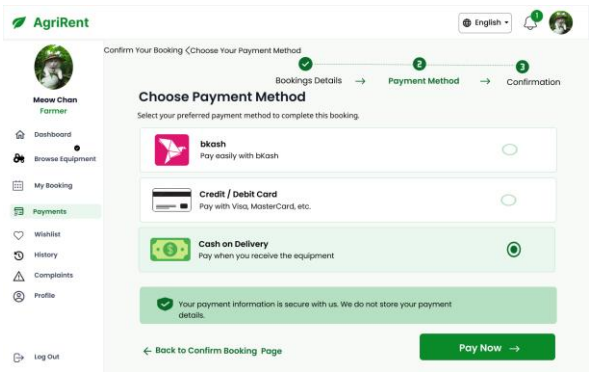


Fig 39:Payment method cash page

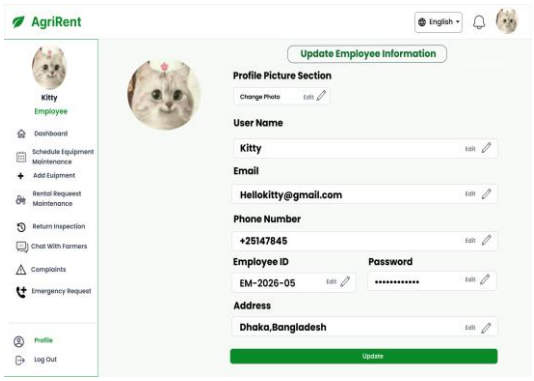


Fig 40:Update employee info page

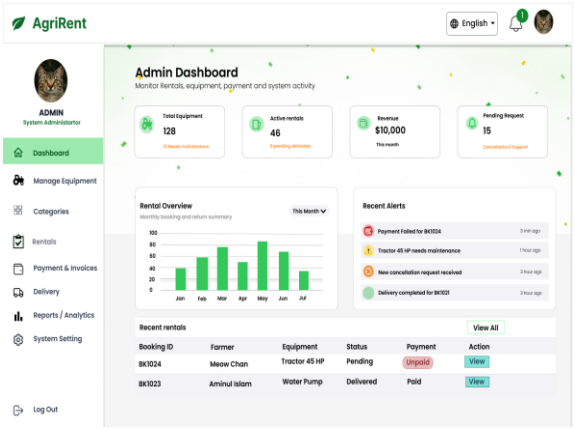


Fig 41: Admin dashboard page

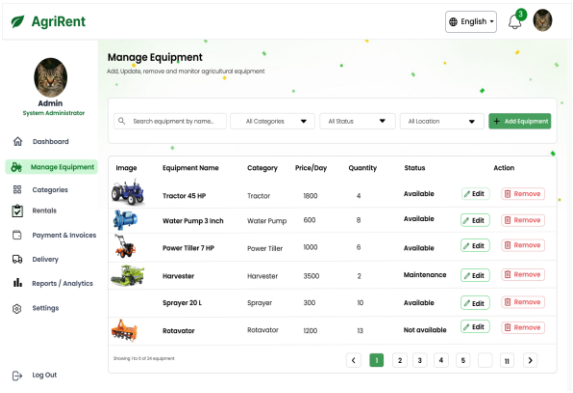


Fig 42:Manage equipment page

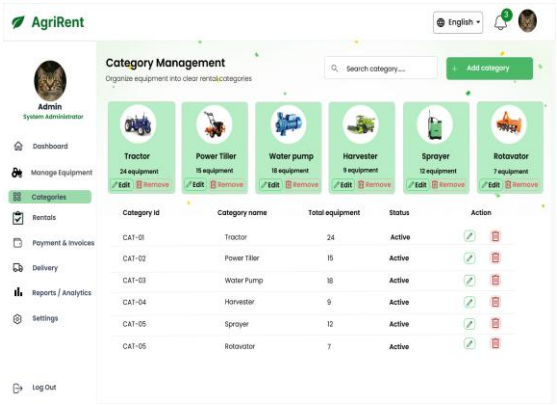


Fig 43: Category manage page

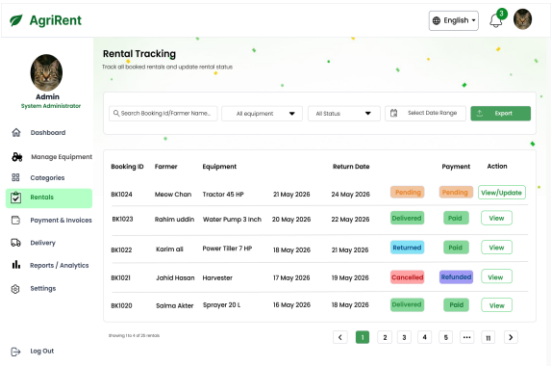


Fig 44:Rental tracking page

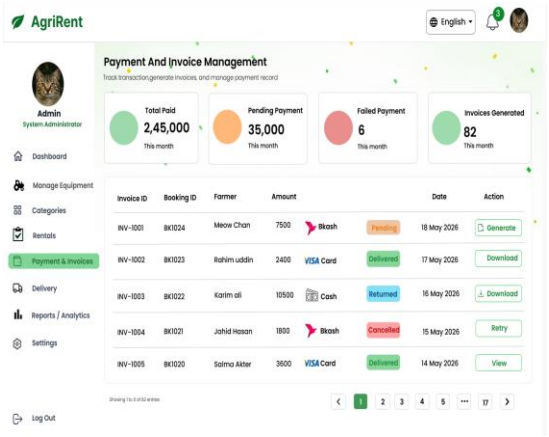


Fig 45: Payment and invoicepage

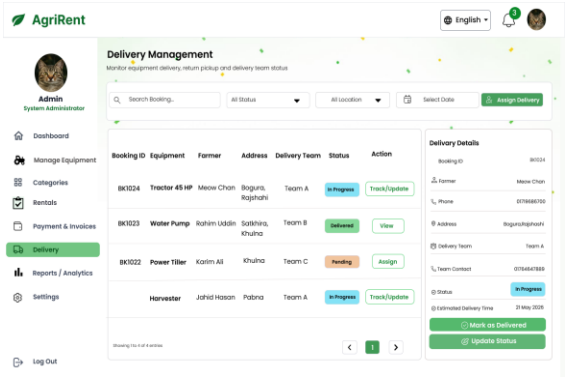


Fig 46: Delivery management page

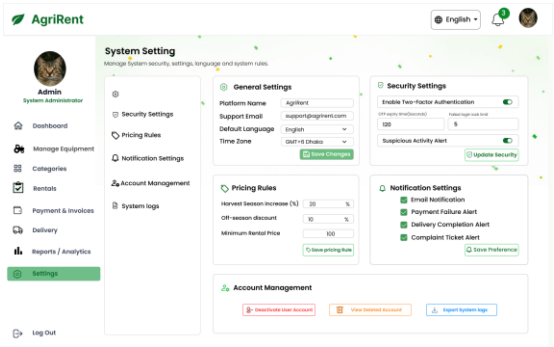


Fig 47: System setting page

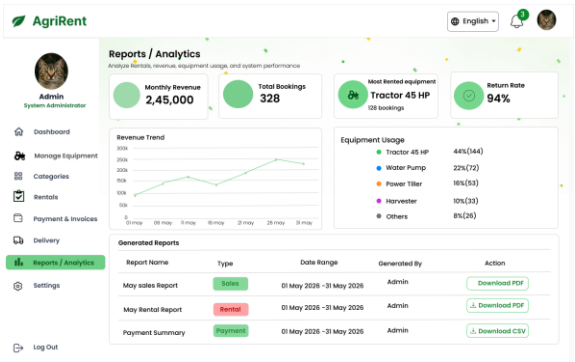


Fig 48: Analytic page

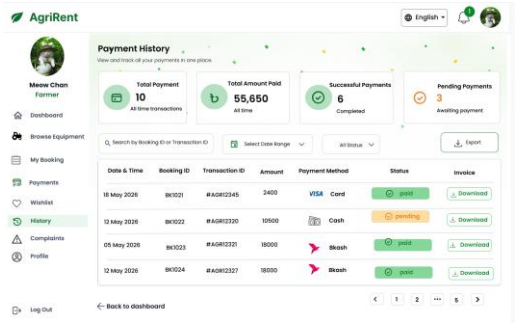


Fig 49: Payment History page

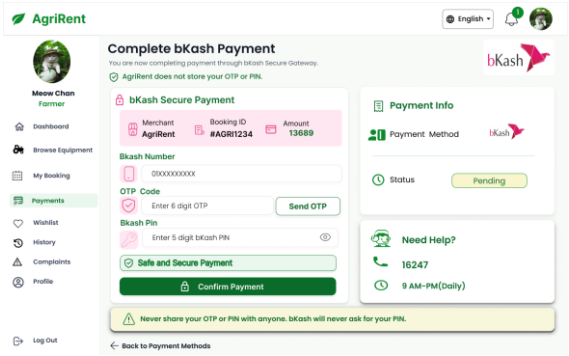


Fig 50: Bkash payment method page

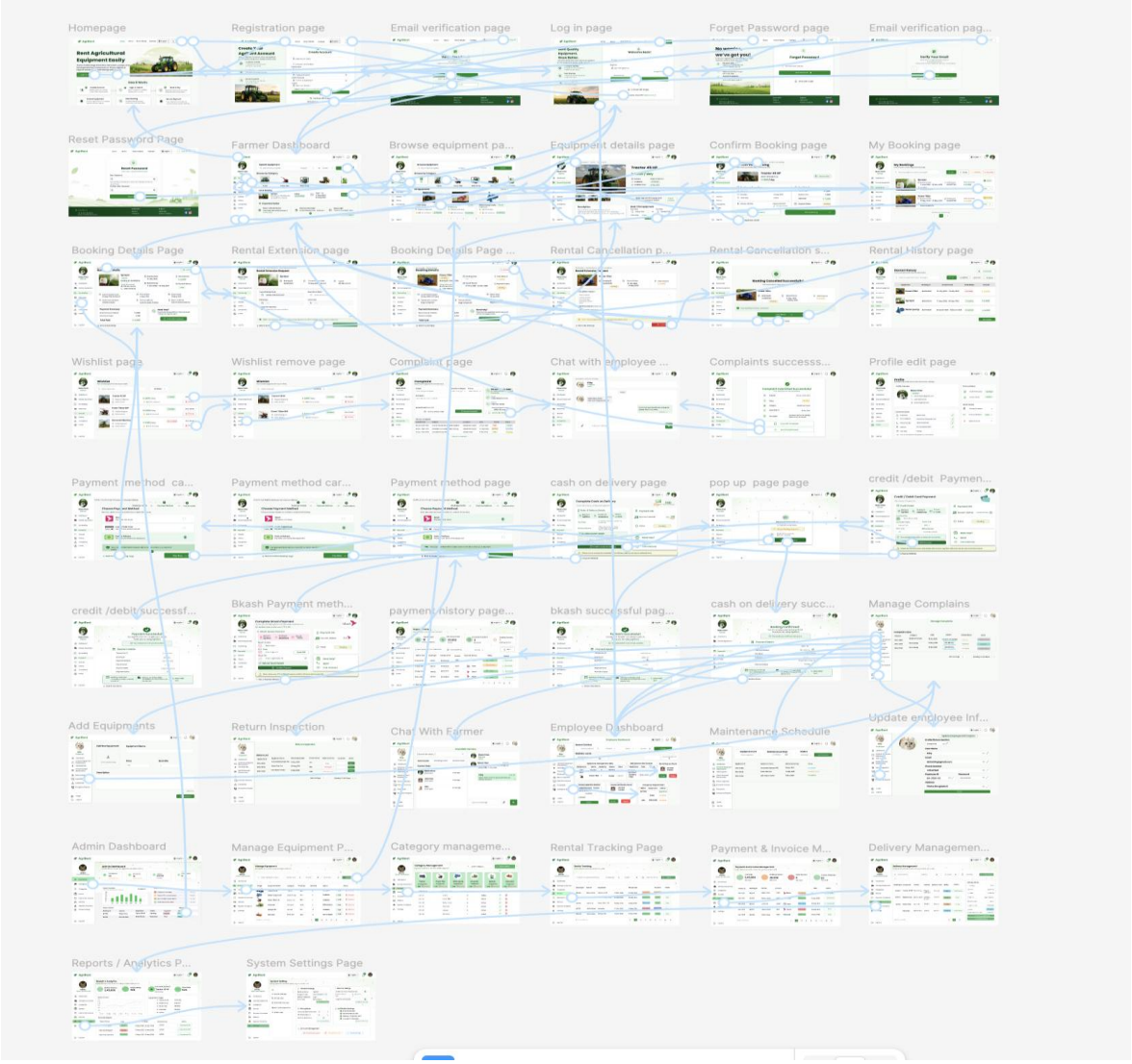
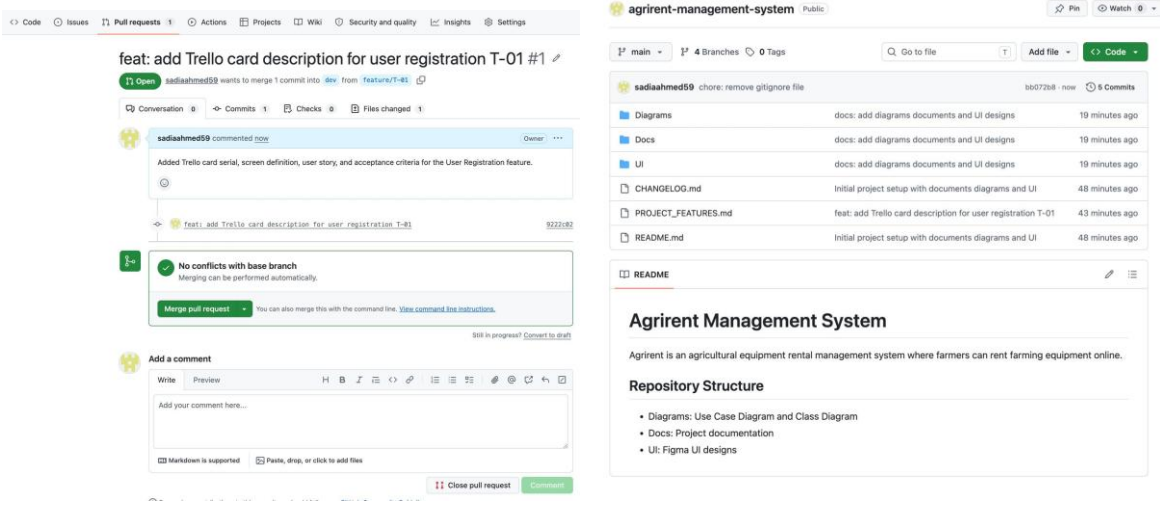


Fig: UI Prototype flow

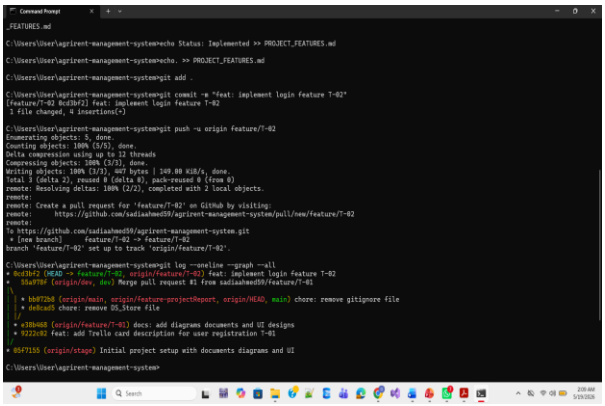
Figma link: https://www.figma.com/design/92rjWx3kj4qrw5pvTFe5jF/Agricultural-Equipment-Rental-And-Management-System?node-id=0-1&t=uvIhS6bg7S890HzH-1

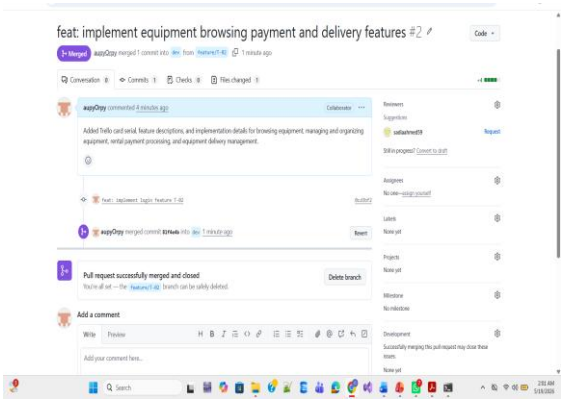
5. GIT WORKFLOW

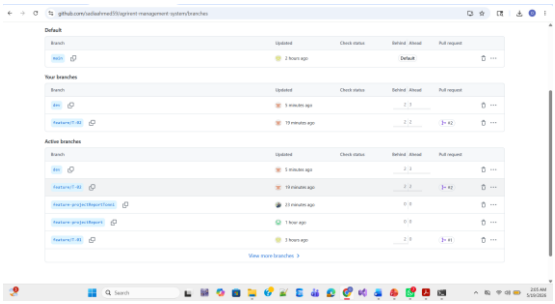




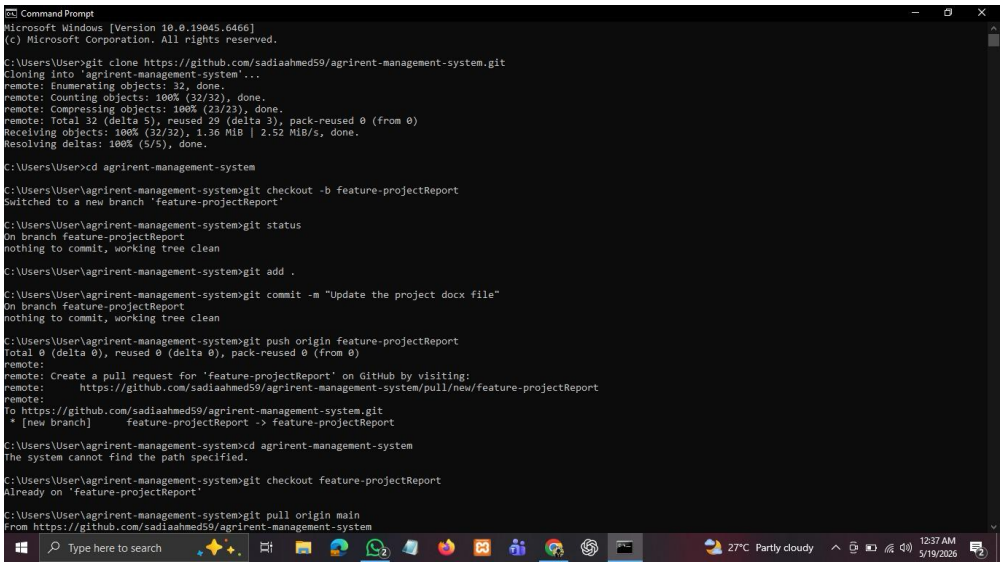
Sadia Ahmed



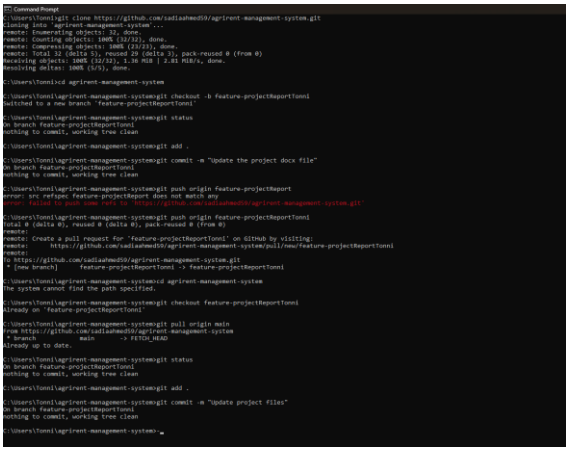


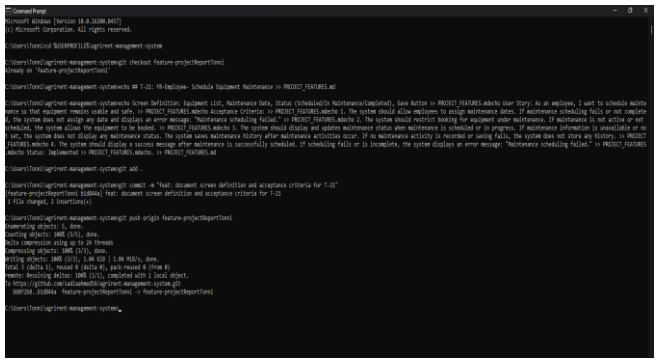


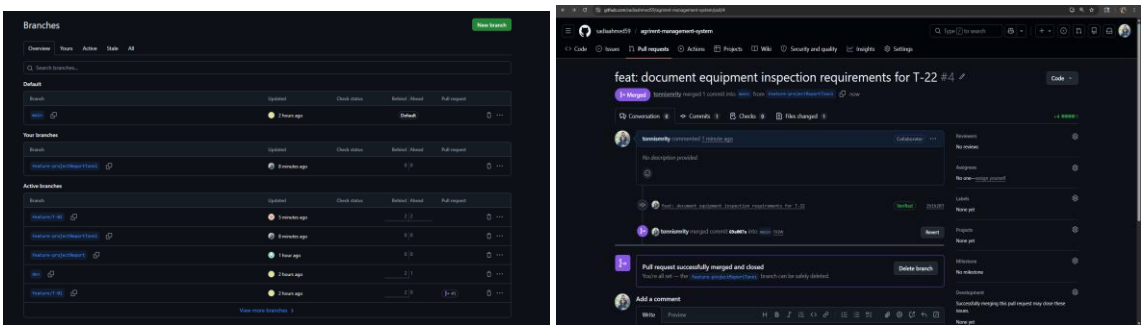
Orpyta karmokar Aupy



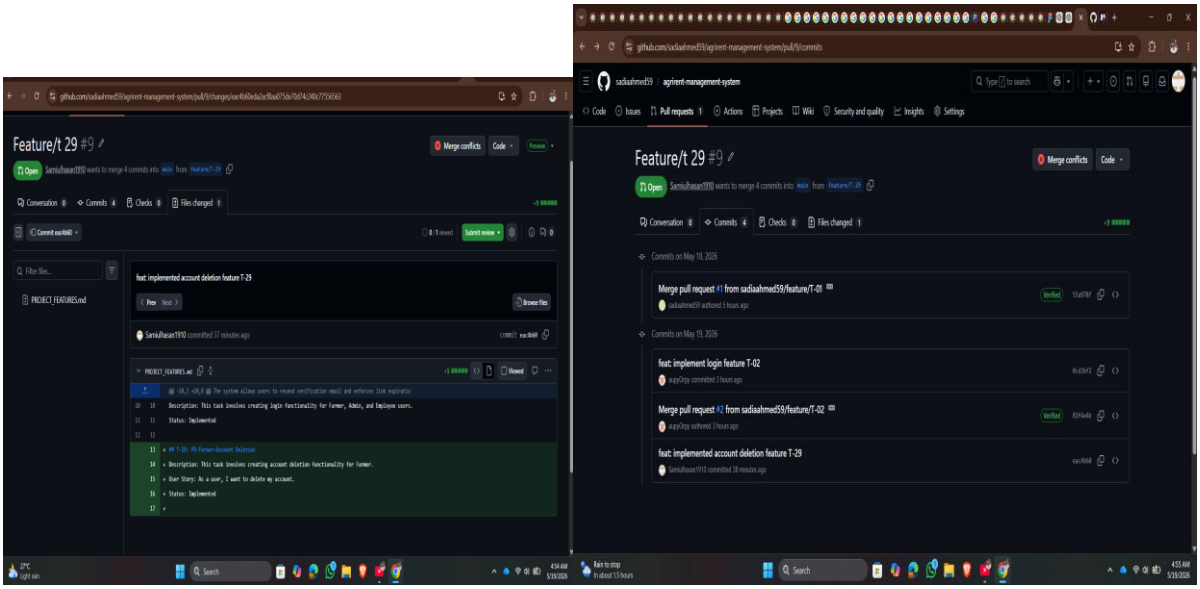
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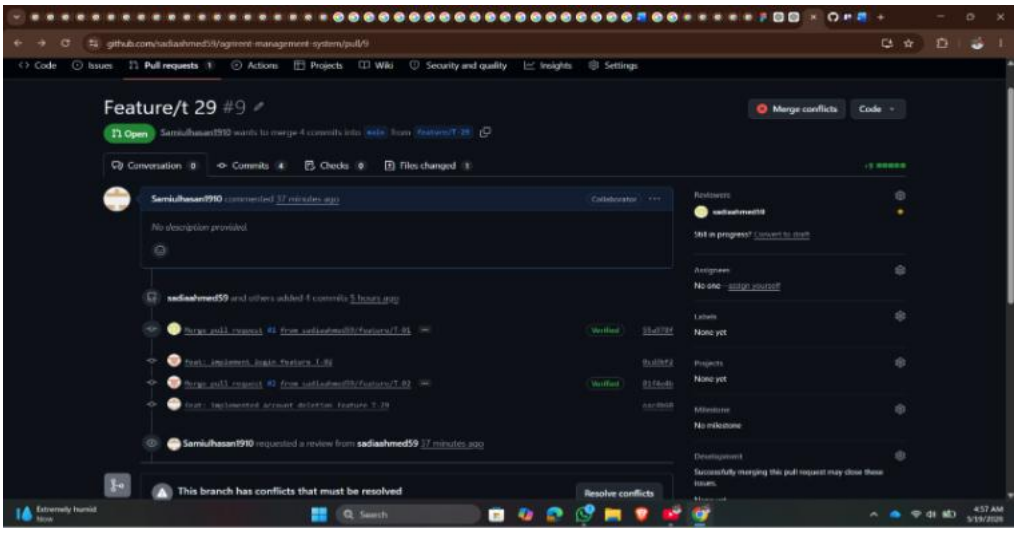






Tanni Mandal





Samiul Hasan

6. SOFTWARE TESTING

Testing Methodology

To ensure the quality, reliability, and security of the **Agricultural Equipment Rental and Management System**, several testing methods were applied throughout the development process. The testing phase was planned based on the system requirements defined in the **PRD**, user workflows, and UI designs.

Unit Testing was performed on individual modules such as User Registration, Login, Forgot Password, Profile Management, Product Management, Cart, and Checkout. Each function was tested independently to verify correct logic, validation rules, and expected outputs.

Integration Testing was conducted to verify proper interaction between related modules. This includes OTP verification during registration and password recovery, database validation during login, cart-to-checkout flow, commission calculation during order placement, and notification updates after payment and delivery status changes.

System Testing was carried out on the fully integrated system to ensure that all modules work together correctly and meet the functional and non-functional requirements specified in the SRS.

User Acceptance Testing (UAT) was considered to confirm that the system meets real-world user expectations. Representative users validated key features such as product browsing, ordering, payment, and delivery tracking.

Security Testing focused on password encryption, OTP authentication, session handling, role-based access control, and protection against common vulnerabilities such as unauthorized access.

Finally, **UI Testing** was performed by comparing implemented screens with the Figma designs to ensure consistency, usability, and responsiveness across different screen sizes.

Project Name: Agricultural Equipment Rental and Management System	Test Designed by:
Test Case ID: TC_AUTH_01	Test Designed date: 10-4-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: User Authentication	Test Execution date:
Test Title: Verify user authentication functionality with valid and invalid scenarios	
Description: Test user authentication functionality including registration, login, forgot password, email verification, account activation, password reset, and profile management.	
Precondition: User has internet access and authentication services are active.	
Dependencies: Database connectivity, Authentication service, Email service, Verification service, File upload service	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open authentication modules 2. Submit invalid/empty registration data 3. Submit valid registration form 4. Attempt login with invalid/unverified credentials 5. Verify account and login successfully 6. Perform forgot password process 7. Update profile information	1. /register, /login, /forgot-password, /profile 2. Blank fields, invalid email, weak password, no user type 3. farmer01@gmail.com / Farmer@123 4. Wrong credentials / Unverified account 5. Valid verification link and credentials 6. Registered email and reset password 7. Updated profile data	• Registration, Login, Forgot Password, and Profile pages load successfully.• Required fields, buttons, and options are visible and accessible. • ystem validates required fields and displays appropriate validation messages.• Invalid email, weak password, mismatched password, and missing user type are rejected.• Signup process is blocked for invalid inputs. • User account is created with inactive status.• Verification email/link is generated and sent successfully.• Show: “Please verify your email”. • System blocks invalid login attempts.• Show: “Email or password does not match!”.• Show: “Please verify your email before logging in”.• Account is temporarily locked after multiple failed attempts. • Valid verification link activates account successfully.• User can login securely using hashed password validation.• User is redirected to dashboard after successful login.• Suspicious login activity triggers notification. • Password reset link is sent successfully.• Invalid/expired reset link is rejected.• New password validation is enforced.• Password is updated successfully after valid reset process. • Profile information loads and updates successfully.• Updated information is displayed immediately.• Appropriate error messages are shown for failed load/update actions.• Show: “Profile updated successfully.”		

Project Name: Agricultural Equipment Rental and Management System	Test Designed by:

Test Case ID: TC_AE_02				Test Designed date: 10-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Add Equipment				Test Execution date:
Test Title: Verify equipment adding functionality				
Description: Test equipment adding functionality including validation, image upload, price validation, and successful submission				
Precondition: Employee must be logged into the system				
Dependencies: Database connectivity, File upload service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Add Equipment page 2. Submit empty form 3. Submit form without image 4. Enter invalid rental price 5. Enter zero/negative quantity 6. Submit valid equipment details 7. View equipment on booking page 8. Confirm rental booking	1. /add-equipment 2. Blank fields 3. No image uploaded 4. Price: abc 5. Quantity: 0 or -5 6. Valid equipment data 7. Added equipment 8. Click "Rent"	• Add Equipment form loads successfully.• Required fields and Add button are visible. • Required field validation messages are displayed.• Equipment is not added. • Show image upload error message.• Submission is blocked. • Show invalid price error message.• Only numeric values are accepted. • Show quantity validation error.• Quantity must be positive. • Equipment is added successfully.• Show: "Equipment added successfully". • Rental price is displayed clearly on booking page.• Equipment details are visible. • Rental confirmation page displays equipment details and price correctly.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_BE_03				Test Designed date: 10-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Browse Equipment				Test Execution date:
Test Title: Verify equipment browsing, searching, and filtering functionality				
Description: Test equipment browsing functionality including equipment display, search, filter, availability status, and navigation				
Precondition: Equipment records must exist in the system				
Dependencies: Database connectivity				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Browse Equipment page 2. Open page with no equipment 3. Search equipment 4. Apply filter options 5. Search/filter with no matching result 6. Check availability status	1. /equipment-list 2. No equipment data 3. Search: Tractor 4. Type, Price, Location 5. Invalid filter/search 6. Equipment availability 7. Click "View Details"	• Equipment list loads successfully.• Name, image, rental price, and availability are displayed. • Show: "No equipment available". • Matching equipment is displayed dynamically.• Equipment list updates automatically. • Filtered equipment list is displayed correctly.• Search/filter updates without page reload. • Show: "No equipment found". • Equipment status shows as "Available" or "Not Available". • User is redirected to equipment details page successfully.		

7. Open equipment details				

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_ME_04				Test Designed date: 13-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Manage Equipment				Test Execution date:
Test Title: Verify equipment management functionality				
Description: Test equipment management functionality including equipment display, add, update, and delete operations				
Precondition: Admin must be logged into the system				
Dependencies: Database connectivity, File upload service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open equipment list page 2. Fail to load equipment list 3. Click Add Equipment button 4. Submit invalid equipment data 5. Submit valid equipment data 6. Fail to save equipment 7. Update equipment details 8. Submit invalid update 9. Remove equipment 10. Fail to delete equipment	1. /manage-equipment 2. Server/DB issue 3. Add action 4. Empty/invalid fields 5. Valid equipment details 6. Server/DB issue 7. Updated equipment data 8. Invalid updated data 9. Select equipment and click Remove 10. Server/DB issue	• Equipment list loads successfully.• Name, price, image, and availability are displayed. • Show: “Failed to load equipment list. Please try again.” • Equipment form opens successfully.• Required fields are visible. • Show: “All fields are required and must be valid.” • Equipment is saved successfully.• New equipment appears in the list. • Show: “Failed to save equipment. Please try again.” • Equipment information is updated successfully.• Updated data appears in equipment list. • Show: “Failed to update equipment. Please ensure all fields are valid and try again.” • Equipment is deleted successfully.• Deleted equipment no longer appears in the list. • Show: “Failed to delete equipment.”• Equipment list refreshes properly.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_MC_05				Test Designed date: 13-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Manage Categories				Test Execution date:
Test Title: Verify category management functionality				
Description: Test category management functionality including category display, add, update, and delete operations				
Precondition: Admin must be logged into the system				
Dependencies: Database connectivity				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open category list page	1. /manage-categories 2. Server/DB issue	• Category list loads successfully.• Existing categories are displayed properly.		

2. Fail to load category list 3. Click Add Category button 4. Submit empty/duplicate category 5. Submit valid category 6. Fail to save category 7. Update category details 8. Submit invalid category update 9. Delete category 10. Fail to delete category	3. Add action 4. Empty or existing category name 5. Tractor 6. Server/DB issue 7. Updated category name 8. Empty/duplicate name 9. Select category and click Remove 10. Server/DB issue	• Show: “Failed to load categories. Please try again.” • Category form opens successfully.• Input field is accessible. • Show: “Category name cannot be empty or duplicated.” • New category is saved successfully.• Category appears in the list.• Show: “Action completed successfully.” • Show: “Failed to save category. Please try again.” • Category information is updated successfully.• Show: “Action completed successfully.” • Show: “Failed to update category. Please ensure the name is valid and try again.” • Category is deleted successfully.• Removed category no longer appears in the list.• Show: “Action completed successfully.” • Show: “Failed to delete category.”• Category list refreshes properly.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_BR_06				Test Designed date: 13-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Booking Equipment for Rental				Test Execution date:
Test Title: Verify equipment booking and rental functionality				
Description: Test equipment booking functionality including rental duration, booking confirmation, price calculation, email confirmation, and rental history				
Precondition: Farmer must be logged into the system and equipment must be available				
Dependencies: Database connectivity, Email service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open equipment booking page 2. Select equipment and rental duration 3. Fail booking process 4. Select rental period 5. Fail price calculation 6. Complete booking successfully 7. Fail sending confirmation email 8. Save booking details 9. Fail saving booking details	1. /booking 2. Valid booking details 3. Invalid/Incomplete booking 4. Rental duration selected 5. Invalid rental input 6. Valid booking information 7. Email service issue 8. Valid booking record 9. Server/DB issue	• Equipment details and rental duration options are displayed successfully. • Booking is confirmed successfully.• Booking details are validated properly. • Show: “Booking could not be completed. Please try again.”• User can retry booking process. • Total rental price is displayed clearly.• Price calculation updates correctly. • Show: “Unable to calculate price. Please try again.”• User can reselect rental period. • Confirmation email with rental details is sent successfully. • Show: “Booking confirmed, but email could not be sent.” • Booking details are saved successfully.• Booking appears in rental history.		

		• Show: “Failed to save booking details.”• User can retry saving process.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_MR_07				Test Designed date: 16-4-2026
Test Priority (Low, Medium, High): HIgh				Test Executed by:
Module Name: Manage & Track Rentals				Test Execution date:
Test Title: Verify rental management and tracking functionality				
Description: Test rental management functionality including rental list display, rental status update, and status synchronization				
Precondition: Admin must be logged into the system and rental records must exist				
Dependencies: Database connectivity				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open rental list page 2. Fail to load rental list 3. Update rental status 4. Fail rental status update 5. Check rental status display 6. Fail status synchronization	1. /manage-rentals 2. Server/DB issue 3. Status: Delivered/Returned 4. Server/DB issue 5. Pending / Delivered / Returned 6. Status mismatch issue	• Rental list loads successfully.• Equipment name, rental status, and user details are displayed. • Show: “Failed to load rental list. Please try again.”• Admin can refresh/reload the list. • Rental status is updated successfully.• Latest status is displayed correctly. • Show: “Failed to update rental status.”• Previous status remains unchanged.• Admin can retry update process. • Correct rental status is displayed for each rental.• Status synchronization is maintained. • Show: “Unable to display updated status.”• System reloads latest status from database.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_RE_08				Test Designed date: 16-4-2026
Test Priority (Low, Medium, High): Medium				Test Executed by:
Module Name: Rental Extension				Test Execution date:
Test Title: Verify rental extension functionality				
Description: Test rental extension functionality including extension request, availability checking, updated pricing, payment confirmation, and notification				
Precondition: Farmer must be logged into the system and an active rental must exist				
Dependencies: Database connectivity, Payment service, Notification service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Request rental extension before expiry 2. Request extension after expiry	1. Valid extension request 2. Expired rental 3. Requested extended period	• Extension request is processed successfully.• Extension option is available before expiry date. • Show: “Extension request cannot be processed. Please try again before the rental expires.”		

3. Check equipment availability 4. Request unavailable extension period 5. Calculate updated rental details 6. Fail rental calculation 7. Confirm extension with payment/approval 8. Fail extension approval/payment 9. Save updated rental details 10. Fail saving updated details 11. Send extension notification 12. Fail sending notification	4. Equipment unavailable 5. Extended rental duration 6. Invalid/missing data 7. Valid payment/approval 8. Payment/approval issue 9. Valid extension data 10. Server/DB issue 11. Successful extension 12. Notification service issue	• Equipment availability is checked successfully.• Extension proceeds if equipment is available. • Show: "Equipment is not available for the selected extended period." • Updated rental duration and total price are displayed clearly. • Show: "Failed to calculate updated rental details. Please try again." • Extension request is confirmed successfully. • Show: "Extension request failed. Please complete the payment or try again." • Updated rental details are saved successfully. • Show: "Failed to update rental details. Please try again." • Confirmation notification is sent successfully. • Show: "Extension successful but failed to send confirmation notification."		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_MP_09				Test Designed date: 16-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Making Payments for Rentals				Test Execution date:
Test Title: Verify rental payment functionality				
Description: Test payment functionality including payment methods, transaction confirmation, coupon application, and receipt email				
Precondition: Farmer must be logged into the system and a booking must exist				
Dependencies: Database connectivity, Payment gateway, Email service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select payment method 2. Complete payment transaction 3. Send payment receipt email 4. Apply coupon code	1. Card / Mobile Wallet 2. Valid / Invalid payment details 3. Successful payment 4. Valid / Invalid coupon	• System provides multiple payment options including credit cards and mobile wallets.• If selected payment method is unavailable, show: "Selected payment method is not available."• User can choose an alternative payment option. • System confirms payment after successful transaction.• If transaction fails, payment process is cancelled.• No amount is deducted.• Show: "Payment failed. Please try again." • System sends receipt email with transaction details after successful payment.• If email sending fails, receipt is stored successfully.• Show: "Payment successful, but receipt email could not be sent." • System applies valid coupon and reduces total price.• If coupon is invalid or expired, show: "Invalid or expired coupon code."• Original price remains unchanged.		

Project Name: Agricultural Equipment Rental and Management System	Test Designed by:
Test Case ID: TC_PM_10	Test Designed date: 19-4-2026

Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Tracking & Managing Payment Activities				Test Execution date:
Test Title: Test payment management functionality including payment status, invoice generation, payment history, notifications, and export operations				
Description: Test payment management functionality including payment status, invoice generation, payment history, notifications, and export operations				
Precondition: Admin must be logged into the system and payment records must exist				
Dependencies: Database connectivity, Invoice service, Notification service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. View payment status 2. Generate invoice 3. View payment history 4. Handle payment failure notification 5. Export payment data	1. Rental payment records 2. Valid rental payment 3. Transaction records 4. Failed payment transaction 5. Payment history / Invoice export	• System displays payment status for each rental.• If payment data fails to load, show: "Unable to load payment status."• Admin can refresh/reload payment data. • System generates invoice with all payment details.• If invoice generation fails, show: "Failed to generate invoice."• Admin can retry invoice generation. • System displays payment history with transaction details.• If payment history fails to load, show: "Failed to load payment history."• Admin can reload payment history. • System notifies admin with rental and user details.• If notification delivery fails, system logs the failure.• Show: "Payment failure notification could not be delivered." • System exports payment history and invoices successfully.• If export process fails, show: "Failed to export payment data."• Admin can retry export process.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_DP_11				Test Designed date: 19-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Managing Equipment Delivery Process				Test Execution date:
Test Title: Verify delivery management and tracking functionality				
Description: Test delivery management functionality including delivery status, tracking progress, contact details, status updates, and notifications				
Precondition: Admin must be logged into the system and delivery records must exist				
Dependencies: Database connectivity, Notification service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. View delivery status 2. Track delivery progress 3. View delivery team contacts 4. Update delivery status 5. Send delivery	1. Delivery records 2. Rental delivery tracking 3. Contact information 4. Status: In Progress / Delivered 5. Completed delivery	• System displays delivery status such as "In Progress" and "Delivered".• If delivery data fails to load, show: "Unable to load delivery status."• Admin can refresh delivery data. • System displays delivery team progress for each rental.• If tracking data fails to load, show: "Failed to load delivery progress."• Admin can reload tracking information. • System displays delivery team phone and email details.• If contact details fail to load, show: "Unable to load contact details."• Admin can refresh contact information. • System updates delivery status successfully.• If update process fails, previous status remains unchanged.• Show:		

completion notification		"Failed to update delivery status."• Admin can retry status update. • System sends notification after delivery completion.• If notification fails, system logs the failure.• Show: "Delivery notification could not be sent."		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_SEM_12				Test Designed date: 19-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Smart Equipment Matching				Test Execution date:
Test Title: Verify smart equipment suggestion functionality				
Description: Test smart equipment matching functionality including input validation, dynamic suggestions, ranking, and booking action				
Precondition: Farmer must be logged into the system and equipment mapping data must exist				
Dependencies: Database connectivity, Recommendation engine				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Submit valid inputs 2. Submit empty/invalid farm size 3. Select unavailable crop type 4. Skip season selection 5. Change input values dynamically 6. Match multiple equipment 7. Select suggested equipment	1. Farm Size, Crop Type, Season 2. Farm Size: Blank 3. Invalid crop type 4. Season: None 5. Update crop/season/farm size 6. Multiple matching equipment 7. Valid equipment selection	• System displays relevant equipment suggestions based on predefined mapping rules. • Show: "Farm size is required".• Suggestion list remains disabled. • Show: "No matching equipment found". • Suggestion list remains inactive. • System refreshes equipment suggestions dynamically without page reload. • System ranks and displays top relevant equipment options. • User can proceed to booking/request action successfully.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_DP_13				Test Designed date: 22-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Dynamic Pricing				Test Execution date:
Test Title: Verify dynamic pricing functionality based on seasonal conditions				
Description: Test dynamic pricing functionality including seasonal adjustment, rule updates, validation, and audit logging				
Precondition: Admin pricing rules and equipment base price must exist				
Dependencies: Database connectivity, Pricing engine				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select Harvest season 2. Select Off-season 3. Calculate price without base price	1. Season: Harvest 2. Season: Off-season 3. Base Price: Null 4. Invalid season input	• System increases equipment price based on predefined rule (e.g., +20%). • System decreases equipment price based on predefined rule (e.g., -10%).		

4. Select invalid/unknown season 5. Update pricing rules 6. Generate final calculated price 7. Change equipment price	5. Admin updates pricing configuration 6. Calculated final price 7. Seasonal price update	• System prevents price calculation.• Error message is displayed. • System ignores seasonal adjustment.• Base price is displayed unchanged. • Updated pricing rules are applied instantly to relevant equipment. • System ensures final price is never negative or zero. • System logs all price changes for audit purposes.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_EER_14				Test Designed date: 22-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Emergency Equipment Request				Test Execution date:
Test Title: Verify emergency equipment request and nearest equipment matching functionality				
Description: Test emergency request functionality including location detection, nearest equipment matching, notification, and retry process				
Precondition: Farmer must be logged into the system and equipment/location data must exist				
Dependencies: Database connectivity, GPS/Location service, Notification service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Send emergency request with location enabled 2. Multiple nearby equipment available 3. Send request with location disabled 4. Search with no nearby equipment 5. Submit successful emergency request 6. Notify admin/employee 7. Fail emergency request process	1. GPS Enabled 2. Multiple equipment locations 3. GPS Disabled 4. No equipment in radius 5. Valid emergency request 6. Successful request 7. Network/System issue	• System identifies nearest available equipment automatically.• Emergency request is assigned successfully. • System selects the closest equipment using distance algorithm. • System prompts user to enable GPS/location services. • Show: “No equipment available nearby”. • User receives instant confirmation message. • System sends notification to admin/employee for approval or monitoring. • System provides retry option for failed request.		

Project Name: Agricultural Equipment Rental and Management System	Test Designed by:
Test Case ID: TC_AD_15	Test Designed date: 22-4-2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Analytics Dashboard	Test Execution date:
Test Title: Verify analytics dashboard and filtering functionality	
Description: Test analytics dashboard functionality including graphs, filtering, validation, and data refresh	
Precondition: Admin must be logged into the system and analytics data must exist	

Dependencies: Database connectivity, Analytics service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select valid date range 2. Apply equipment filter 3. Select invalid date range 4. Apply filter with no matching data 5. Load analytics dashboard 6. Display analytics graphs 7. Refresh dashboard data	1. Start Date < End Date 2. Selected equipment 3. End Date earlier than Start Date 4. Empty analytics data 5. Dashboard access 6. Revenue/Booking/Usage data 7. Real-time/Periodic refresh	• System displays revenue, bookings, and usage graphs according to selected date range. • System displays filtered analytics data for selected equipment. • System displays validation error for invalid date range. • System displays: "No data available". • Dashboard loads within acceptable time without errors. • Graphs accurately represent stored system data. • System refreshes analytics data successfully in real-time or periodically.		

Project Name: Agricultural Equipment Rental and Management System	Test Designed by:
Test Case ID: TC_EHM_16	Test Designed date: 25-4-2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Equipment Health Monitoring	Test Execution date:
Test Title: Verify equipment health monitoring and maintenance alert functionality	
Description: Test equipment health monitoring functionality including maintenance alerts, status updates, usage logs, and booking restriction	
Precondition: Equipment usage and maintenance records must exist in the system	
Dependencies: Database connectivity, Monitoring service	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Exceed equipment usage threshold 2. Mark equipment as recently serviced 3. Detect due maintenance 4. Load incomplete usage data 5. Store maintenance records 6. View maintenance alerts 7. Attempt booking unavailable equipment	1. Usage above predefined limit 2. Updated service record 3. Maintenance due equipment 4. Missing/incomplete usage record 5. Maintenance and usage logs 6. Dashboard alerts 7. Equipment status: "Needs Maintenance"	• System automatically generates maintenance alert. • System sets equipment status as "Good". • System updates equipment status to "Needs Maintenance". • System displays warning message.• System flags incomplete record. • System stores maintenance history and usage logs persistently. • Maintenance alerts are visible to admin/employee in dashboard. • System prevents booking of equipment marked as "Needs Maintenance".		

Project Name: Agricultural Equipment Rental and Management System	Test Designed by:
Test Case ID: TC_MOA_17	Test Designed date: 25-4-2026

Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Multilingual & Offline Access				Test Execution date:
Test Title: Verify multilingual support and offline synchronization functionality				
Description: Test multilingual UI switching, offline access, synchronization, and data consistency functionality				
Precondition: Farmer must be logged into the system				
Dependencies: Database connectivity, Local storage, Synchronization service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select Bangla language 2. Select English language 3. Enable offline mode 4. Restore internet connection 5. Fail synchronization process 6. Check synchronization status 7. Perform offline-to-online transition	1. Language: Bangla 2. Language: English 3. Internet disconnected 4. Internet connected 5. Sync service/network issue 6. Sync operation 7. Offline data update	• Entire UI is translated to Bangla consistently. • System switches back to English UI successfully. • User can access basic features using locally stored data. • System automatically synchronizes offline data with server. • System displays synchronization error message.• Retry option is available. • System displays sync status as pending/success/failed. • System ensures no data loss during synchronization process.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_SEM_18				Test Designed date: 25-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Schedule Equipment Maintenance				Test Execution date:
Test Title: Verify equipment maintenance scheduling functionality				
Description: Test maintenance scheduling functionality including maintenance status, booking restriction, and maintenance history				
Precondition: Employee must be logged into the system and equipment records must exist				
Dependencies: Database connectivity				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Schedule maintenance date 2. Attempt booking equipment under maintenance 3. Update maintenance status 4. Complete maintenance scheduling	1. Valid maintenance date 2. Equipment status: In Maintenance 3. Scheduled / In Maintenance 4. Successful scheduling	• System assigns maintenance date successfully.• If scheduling fails, no maintenance date is assigned.• Show: “Maintenance scheduling failed.” • System restricts booking for equipment under maintenance.• If maintenance is not active/scheduled, booking is allowed. • System displays and updates maintenance status correctly.• If maintenance information is unavailable, no maintenance status is displayed.• System saves maintenance history after maintenance activity.• If saving fails, maintenance history is not stored. • System displays success message after scheduling.• If scheduling fails or is incomplete, show: “Maintenance scheduling failed.”		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_IRE_19				Test Designed date: 28-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Inspect Returned Equipment				Test Execution date:
Test Title: Verify returned equipment inspection functionality				
Description: Test returned equipment inspection functionality including condition selection, comments, availability update, and inspection record saving				
Precondition: Employee must be logged into the system and returned equipment records must exist				
Dependencies: Database connectivity				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select equipment condition 2. Add inspection comments 3. Mark equipment as damaged 4. Save inspection data 5. Submit inspection process	1. Good / Damaged / Needs Service 2. Comment text 3. Condition: Damaged 4. Inspection details 5. Completed inspection form	• System records selected condition successfully.• If condition is not selected or process fails, condition is not recorded.• Show: “Condition selection failed.” • System stores additional notes successfully.• If comment saving fails or field is empty, notes are not stored. • System marks damaged equipment as unavailable.• If equipment is not damaged, it remains available for use. • System stores inspection information successfully.• If saving fails, inspection information is not stored. • System displays confirmation message after successful submission.• If submission fails or is incomplete, show: “Confirmation failed.”		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_RRC_20				Test Designed date: 28-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Request Rental Cancellation				Test Execution date:
Test Title: Verify rental cancellation functionality				
Description: Test rental cancellation functionality including cancellation validation, reason submission, booking status update, notification, and booking history				
Precondition: Farmer must be logged into the system and active booking records must exist				
Dependencies: Database connectivity, Notification service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Cancel booking before delivery 2. Submit cancellation without reason 3. Submit valid cancellation request 4. Send cancellation notification 5. Save cancelled booking history	1. Booking status: Pending 2. Empty reason field 3. Valid cancellation reason 4. Successful cancellation 5. Cancelled booking record	• System allows cancellation before delivery.• If delivery is completed or in progress, show: “Cancellation not allowed after delivery.” • System requires cancellation reason.• If reason is missing or submission fails, cancellation is not processed.• Show: “Cancellation reason required.” • System updates booking status to “Cancelled”.• If cancellation fails, booking status remains unchanged. • System sends cancellation notification to admin.• If notification fails, admin does not receive alert.		

		• System keeps cancelled booking in history.• If saving fails, cancelled booking is not stored in history.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_ERR_21				Test Designed date: 28-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Submit Equipment Review & Rating				Test Execution date:
Test Title: Verify equipment review and rating submission functionality				
Description: Test review and rating functionality including rating selection, review submission, data storage, and confirmation message				
Precondition: Farmer must be logged into the system and completed rental records must exist				
Dependencies: Database connectivity				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Submit rating selection 2. Submit review text 3. Save review and rating 4. Complete review submission	1. Rating: 1–5 stars 2. Review description 3. Valid rating and review 4. Successful review submission	• System allows rating selection between 1 to 5 stars.• If no rating is selected or submission fails, rating is not submitted.• Show: "Rating selection required." • System allows user to write review in text box.• If review field is empty or submission fails, review is not submitted.• Show: "Review description cannot be empty." • System stores rating and review successfully.• If saving fails, review is not stored.• Show: "Review submission failed." • System displays confirmation message: "Review submission successful."		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_CST_22				Test Designed date: 30-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Manage Complaint & Support Tickets				Test Execution date:
Test Title: Verify complaint submission and support ticket management functionality				
Description: Test complaint management functionality including category selection, description validation, ticket generation, status update, and status tracking				
Precondition: User must be logged into the system				
Dependencies: Database connectivity, Ticket management service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Submit complaint category 2. Submit complaint description 3. Generate support ticket 4. Update complaint status 5. View complaint	1. Selected / Empty category 2. Complaint details 3. Valid complaint submission 4. Admin status update 5. Updated complaint record	• System allows complaint category selection.• If no category is selected or submission fails, complaint is not submitted.• Show: "Complaint category selection failed." • System requires description before submission.• If description is empty or submission fails, complaint is not submitted.• Show: "Description cannot be empty." • System generates unique ticket ID successfully.• If ticket generation fails, complaint record is not created.• Show: "Ticket ID generation failed."		

status update		• System allows admin to update complaint status.• If update fails or is incomplete, complaint status remains unchanged.• Show: "Complaint status update failed." • System displays updated complaint status to user.• If status cannot be retrieved, no updated status information is displayed.		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_RHR_23				Test Designed date: 30-4-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: View & Download Rental History Report				Test Execution date:
Test Title: Verify rental history viewing and report download functionality				
Description: Test rental history report functionality including rental display, filtering, report generation, report validation, and PDF download process.				
Precondition: Farmer must be logged into the system and rental history records must exist.				
Dependencies: Database connectivity, Report generation service, PDF export service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. View rental history list 2. Apply rental filters 3. Download rental history report 4. Verify report content 5. Validate report data accuracy	1. Rental records 2. Date range / Status filter 3. Click Download (PDF) 4. Equipment name, duration, price 5. System rental records	• System displays all rental records successfully.• If rental data cannot be retrieved, show: "Unable to load rental data." • System applies filters correctly based on date and status.• If filter process fails or invalid criteria are used, show: "Filter operation failed." • System generates and downloads report successfully.• If report generation fails, show: "Report generation failed." • Report includes equipment name, rental duration, and price correctly.• If required report data is missing, show: "Incomplete report data." • Report data matches system records accurately.• If mismatch is detected, show: "Report data mismatch detected."		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_VFA_24				Test Designed date: 01-05-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Verify Farmer Account				Test Execution date:
Test Title: Verify farmer account verification and approval functionality				
Description: Test farmer account verification functionality including verification request display, farmer details review, approval/rejection process, booking restriction, verification status, and notification process.				
Precondition: Employee must be logged into the system and farmer verification requests must exist.				
Dependencies: Database connectivity, Document upload service, Notification service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. View verification request list 2. View farmer details and documents 3. Approve farmer account 4. Reject farmer account 5. Attempt booking with pending/rejected account 6. View farmer verification status 7. Send verification notification	1. Pending farmer accounts 2. Farmer profile & uploaded document 3. Valid farmer information 4. Invalid/incomplete information 5. Pending / Rejected farmer account 6. Approved / Rejected status 7. Approval / Rejection action	• System displays list of newly registered farmers waiting for verification.• If verification requests cannot be retrieved, show: "Unable to load verification requests." • System displays farmer details and submitted documents successfully.• If details/documents fail to load, show: "Unable to load farmer details." • System approves farmer account successfully.• If approval process fails, account status remains unchanged.• Show: "User verification failed. Please try again." • System rejects farmer account successfully.• If rejection process fails, account status remains unchanged.• Show: "User verification failed. Please try again." • Approved farmers can book equipment successfully.• Pending/rejected farmers cannot book equipment.• Show: "Account verification required before booking." • System displays updated verification status to farmer.• If status retrieval fails, show: "Unable to load verification status." • System sends notification to farmer after approval/rejection.• If notification fails, verification result is saved successfully.• Show: "Verification completed, but notification could not be sent."		

Project Name: Agricultural Equipment Rental and Management System				Test Designed by:
Test Case ID: TC_ADL_25				Test Designed date: 01-05-2026
Test Priority (Low, Medium, High): High				Test Executed by:
Module Name: Account Deletion				Test Execution date:
Test Title: Verify account deletion functionality				
Description: Test account deletion functionality including deletion request, confirmation process, authentication validation, data removal, and login restriction after deletion.				
Precondition: Farmer account must exist in the system.				
Dependencies: Database connectivity, Authentication service				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Request account deletion 2. Confirm/cancel deletion action 3. Attempt deletion without login 4. Delete account successfully	1. Delete account option 2. Confirm / Cancel action 3. Unauthenticated user 4. Valid authenticated account	• System allows user to request account deletion successfully.• If request fails, show: "Failed to process account deletion request. Please try again." • System asks for confirmation before deleting account.• If user cancels the action, account remains unchanged.		

5. Complete account deletion process 6. Attempt login with deleted account	5. Deleted account 6. Deleted account credentials	• Only authenticated users can delete account.• Show: “You must be logged in to delete your account.” • System removes or marks user data as inactive successfully.• If deletion process fails, show: “Failed to delete account data. Please try again.” • System displays: “Your account has been deleted successfully.” • System prevents login using deleted account.• Show: “This account no longer exists or has been deactivated.”		

7. CONCLUSION

The Agricultural Equipment Rental and Management System provides a practical and user-friendly solution for farmers who need agricultural equipment for a specific period without purchasing expensive machines. The system allows farmers to browse available equipment, check details, book rentals, make payments, manage bookings, request rental extensions, cancel eligible bookings, submit complaints, chat with employees, and maintain their profile information. It also supports administrators and employees in managing equipment, rental tracking, payments, delivery, complaints, maintenance, inspections, and reports.

This project improves the rental process by making it faster, more organized, and more transparent. Features such as smart equipment suggestions, emergency equipment requests, wishlist, rental history,and multilingual support make the system more useful for real-life farming needs. The system also focuses on security, clear error messages, and easy navigation so that farmers can use it comfortably.

Overall, this system can reduce manual rental problems, improve communication between farmers and service providers, and ensure better management of agricultural equipment. It is a scalable web-based solution that can help modernize agricultural equipment rental services.

Trello link: https://trello.com/b/OmDgi9mf/agricultural-equipment-rental-and-management-system